

Process Characterization Report

A-Mab Drug Substance — Production Bioreactor (Step 3) — PCR-003

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

2-AB, 2-aminobenzamide; ADCC, antibody dependent cellular cytotoxicity; AMV, analytical method validation; ANOVA, analysis of variance; BLA, Biologics License Application; CCD, central composite design; Cpk, process capability index; CPP, critical process parameter; CQA, critical quality attribute; DoE, design of experiments; GPP, general process parameter; HCP, host cell protein; HILIC, hydrophilic interaction liquid chromatography; HMW, high molecular weight; icIEF, imaged capillary isoelectric focusing; KPP, key process parameter; LOQ, limit of quantitation; MSAT, Manufacturing Science and Technology; NOR, normal operating range; PAR, proven acceptable range; pCO₂, dissolved carbon dioxide partial pressure; PPQ, process performance qualification; PRESS, predicted residual error sum of squares; qPCR, quantitative polymerase chain reaction; RMSE, root mean square error; RSM, response surface methodology; SDM, scale down model; SEC, size exclusion chromatography; SOP, standard operating procedure; UPLC, ultra performance liquid chromatography; WC-CPP, well controlled critical process parameter.

Executive summary

The production bioreactor forms the glycan, charge variant and aggregate quality attributes of A-Mab, and this report bounds the culture conditions that set them. The bioreactor studies were executed on a qualified scale down model of the 15,000 L production vessel, under the plan approved as PCP-003 and the risk basis established in RA-001. The study approach follows the lifecycle model for process validation (U.S. Food and Drug Administration 2011) and the enhanced development approach of ICH Q8 and ICH Q11 (International Council for Harmonisation 2009, 2012), and the attribute framework, the criticality levels and the risk ranking method that fixed the study scope are those of the A-Mab case study (CMC Biotech Working Group 2009).

The screening study covered 5 parameters in a two level design of 19 runs. The 4 factors that screening retained then entered a face centred response surface design of 28 runs, and the remaining 4 parameters were assessed one at a time. Screening identified which factors matter. The response surface model built on those factors is the predictive model, and the design space and the proven acceptable ranges reported below rest on it, not on the near saturated screening fit.

The response surface models account for between 91.1% and 97.1% of the variance in the 5 measured responses, at an adjusted R^2 of 0.81 or better. No response shows significant lack of fit against the centre point replicates, and the weakest of the five on predicted R^2 is acidic variants, at 0.60.

Across the characterized ranges, 57 % of an evenly spaced grid of 14,641 operating points meets every in-process criterion the step governs. The characterized ranges contain edges of failure, which is what separates the knowledge space from the design space. The response that rejects most of the design grid is galactosylation, at 14 % of the points.

Commercial scale capability was estimated from a Monte-Carlo simulation of 2,000 batches with every parameter varying inside the normal operating range. The simulated batches meet the drug substance criterion for all 7 attributes the step governs. The tightest capability of the 7 attributes is the one for high mannose, at a Cpk of 1.94, whose simulated mean sits 5.8 standard deviations from the nearer of the two limits.

Of the 9 parameters characterized, 5 are classified as well controlled critical process parameters (WC-CPP) and 3 as key process parameters (KPP). The remaining parameter, basal medium concentration, is a general process parameter (GPP). No parameter at this step is classified as a critical process parameter, because the control system holds each of the quality linked parameters inside ranges the study proved acceptable.

The execution recorded 2 deviations, and both were retained after investigation. Neither changed a parameter classification and neither moved the boundaries of the

operating region (§13). The step does not control host cell protein (HCP) or residual DNA to the drug substance criteria by itself. Host cell protein and residual DNA are both cleared downstream, and that clearance is credited in PCR-005, PCR-007 and PCR-008. The capability reported here is an estimate from a qualified scale down model and not a commercial scale result. Confirmation at scale belongs to Stage 2, and this report rolls up into PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody whose principal mechanism of action is antibody dependent cellular cytotoxicity. That mechanism depends on the N-glycan profile of the Fc region, which makes afucosylation and galactosylation attributes of efficacy as well as of consistency. The antibody is expressed by the A-Mab production cell line and is produced by fed batch culture in the production bioreactor, which is Step 3 of the drug substance process. Downstream of the bioreactor the platform train runs Protein A capture, low pH viral inactivation, cation exchange, anion exchange, small virus retentive filtration and ultrafiltration with diafiltration (CMC Biotech Working Group 2009).

The production bioreactor is the only step of the drug substance process at which product quality attributes are formed. The glycosylation and charge variant distributions of the product are established inside the cell and in the culture fluid, and the platform purification train does not modify them. Culture pH, temperature, dissolved carbon dioxide, osmolality and duration act on the glycosyltransferases of the Golgi and on the chemistry of the culture fluid, and the levels they set are carried into the drug substance. The culture conditions therefore define the design space for the glycan, charge variant and aggregate attributes of A-Mab, and Table 2.1 lists those attributes with the criticality and acceptance criteria of each.

The vessel is operated as a fed batch culture at a working volume of 15,000 L at commercial scale, under SOP-2003. The culture receives nutrient feed on a fixed schedule as a percentage of working volume, and is harvested at the end of the scheduled duration. At the set-point conditions the culture delivers a harvest titre of 4.4 g/L, a peak viable cell density of 13.5×10^6 cells/mL and a final viability of 67 %.

Two of the attributes the bioreactor governs leave the step far above the level the drug substance must meet. Both of those attributes, host cell protein and residual DNA, accumulate in the culture fluid and are removed by the purification train. Therefore the bioreactor sets the burden that train receives, not the released value. The clearance credit for both belongs to Protein A capture (PCR-005), cation exchange (PCR-007) and anion exchange (PCR-008), and no clearance of any kind is claimed for this step.

1.2 Objectives

The study set out to do five things.

- (i) Quantify the relationship between each characterized culture parameter and each quality attribute the step forms.
- (ii) Define a multivariate operating region over the parameters that govern those attributes, at commercial equivalent conditions.
- (iii) Establish proven acceptable ranges for each governed attribute and each response surface factor, and confirm the normal operating ranges against them.
- (iv) Classify every characterized parameter against the decision logic in SOP-4001.
- (v) Justify the ranges that Stage 2 will run and the contribution this step makes to the control strategy.

1.3 Regulatory and scientific basis

The study is a Stage 1 process design activity in the sense of the lifecycle approach to process validation (U.S. Food and Drug Administration 2011). The study record is this report and the plan it executes, PCP-003. Stage 2 confirms at commercial scale what Stage 1 establishes at scale down, and nothing in this report substitutes for that confirmation.

The development approach is the enhanced, quality by design approach of ICH Q8 and ICH Q11 (International Council for Harmonisation 2009, 2012). Under that approach a design space is a multivariate region of parameter settings that has been shown to assure quality, movement inside the region is not a change from a regulatory filing perspective, and the burden the approach places on a report of this kind is to show how the region was arrived at and where it stops. Quality attribute criticality is treated as a continuum rather than as a binary (International Council for Harmonisation 2023), which is how the attribute levels in Table 2.1 were assigned and how the parameter classes in §9 follow from the measured effects.

ICH Q5A is not engaged at this step. Viral safety rests on the low pH hold (PCR-006), anion exchange (PCR-008) and small virus retentive filtration (PCR-009), and the cumulative claim is consolidated in PCMR-001.

1.4 Related documents

The characterization package this report belongs to is listed in Table 1.1, and the controlled procedures and validated methods the study used are given in Table 3.1 (§3.3).

Table 1.1: Related documents.

Docu- ment ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-003	Process Characterization Plan (Production Bioreactor (Step 3))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

The production bioreactor runs on an established humanized IgG1 platform. Three earlier products from that platform, X-Mab, Y-Mab and Z-Mab, were manufactured with the same medium, the same feed strategy and the same vessel design, and the acceptance limits for several of the A-Mab attributes in Table 2.1 were set from X-Mab clinical experience (CMC Biotech Working Group 2009). Platform experience of that depth changes what a characterization study is for, because the mechanisms by which culture conditions move a glycan profile are already known. Consequently the study confirms and bounds those mechanisms for A-Mab instead of discovering them.

Four of those mechanisms were carried into the design as explicit expectations. Galactosylation tracks the activity of the Golgi galactosyltransferase and the availability of its nucleotide sugar donor, both of which decline as a culture ages and as nutrients deplete. High mannose is enzymatic in the same way and reflects how completely the Golgi mannosidases have trimmed the precursor, which makes it sensitive to luminal pH and to temperature. Afucosylation is set by a competition instead, between core fucosyltransferase activity and the rate at which the antibody transits the secretory pathway. Aggregate is the one expectation that is not enzymatic, because antibody that sits longer in an ageing broth aggregates more.

Acidic charge variants arise mainly from the deamidation of asparagine, and deamidation is base catalysed. Dissolved carbon dioxide acidifies the culture fluid, which was expected to slow the reaction and to lower the acidic fraction. Culture temperature was expected to act in the opposite direction through the rate constant.

Prior knowledge does not settle two things, and both of them shaped this study. It does not give the position of the A-Mab operating point relative to the edges of failure for these attributes, because those edges depend on the cell line and on the acceptance limits the product carries. It also says little about interaction, since the platform record is a record of operation at target, not of parameters moved together. For this reason the study of these parameters was designed as a multivariate one over ranges wider than the control ranges, rather than as a confirmation at set-point.

2.2 Quality attributes in scope

For this step, Table 2.1 lists the 7 attributes the production bioreactor governs, with the criticality level and the Tool #1 score behind it. Tool #1 scores an attribute as the product of the impact on safety or efficacy and the uncertainty in that impact, and the

levels run from very low to very high (CMC Biotech Working Group 2009). Attributes at high and very high criticality are treated as critical.

Table 2.1: Quality attributes formed or governed at the production bioreactor.

CQA	Category	Acceptance	Criticality	Tool #1
Afucosylation	glycosylation	2-13 % of glycans	H	60
Galactosylation (total %Gal)	glycosylation	10-40 % (G1+G2 equiv.)	H	48
High Mannose	glycosylation	3-10 % of glycans	VH	80
Aggregates (HMW)	size variant	0-5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18-40 % (CEX/icIEF)	VL	4
Host Cell Protein (HCP)	process impurity	0-100 ng/mg	M-H	36
Residual DNA	process impurity	0-0.001 ng/dose	VL	6

The most consequential of these attributes is high mannose, at very high criticality and a Tool #1 score of 80. High mannose glycans shorten circulating half-life and the uncertainty around that effect is itself high, which is what drives the score. The next two glycan attributes, afucosylation and galactosylation (total %gal), are of high criticality for a different reason: both modulate effector function, and afucosylation does so steeply. The fourth attribute, aggregate, is of high criticality as an immunogenicity risk, and it is the one attribute in this group that later steps also act on (PCR-006, PCR-007).

By contrast, acidic charge variants sit at the other end of the range, at very low criticality and a Tool #1 score of 4. Deamidation at the levels this charge variant reaches has no demonstrated effect on potency, and the acceptance criterion of 18 to 40 % is correspondingly wide. The charge variant is nonetheless characterized here, because it responds to the same culture parameters as the glycans and because an unexplained shift in the attribute would be a signal about the culture.

Host cell protein and residual DNA belong to a different category, as process impurities. Both are formed in the bioreactor and both are cleared downstream, and the criticality levels of M-H and VL reflect the released material, not the harvest. Neither entered the multivariate design. Culture duration drives the harvest burden of both, and the range studied for duration bounds the load the purification train has to handle.

2.3 Risk-based prioritization and parameter selection

RA-001 assigned the study scope by risk ranking and filtering. Every parameter of the step was scored on the potential impact on a quality attribute and on the potential to interact with other parameters, and the combined score decided whether the parameter went into a multivariate design, into a univariate assessment, or into neither (CMC Biotech Working Group 2009; International Council for Harmonisation 2023). The parameter was ranked at the highest level wherever no data or rationale supported a judgement.

Multivariate study was required for 5 parameters: culture pH, culture temperature, dissolved CO₂ (pCO₂), osmolality and culture duration. Each of them acts directly on the enzymology or the solution chemistry that sets the glycan and charge variant profile, and each was expected to interact with at least one of the others. Culture duration is the clearest case, since it sets both the extent of glycan processing and the residence time of secreted antibody in the broth.

The remaining 4 parameters were assessed one at a time: dissolved oxygen, initial viable cell conc., basal medium concentration and nutrient feed-1 volume. All four parameters act on cell growth and productivity, and none of them touches the reactions that set product quality. RA-001 scored them below the multivariate threshold on that basis. Dissolved oxygen and initial viable cell concentration set the growth trajectory, basal medium concentration sets the nutrient reservoir, nutrient feed-1 volume sets the replenishment, and the outcome of the assessment for each of the four is the classification given later (§9).

For the five multivariate parameters, the characterization ranges are wider than the control ranges by a factor of between 1.6 and 2.5. Widening them was deliberate. A range that only spans the control range cannot show where an attribute leaves specification, and cannot support a later request to move within the design space. The ranges in Table 4.1 are the knowledge space of this step, and they extend well beyond any setting the commercial process is expected to use.

3 Materials and methods

3.1 Scale-down model and its qualification

The studies were executed in an instrumented bench scale bioreactor system qualified as a scale down model (SDM) of the 15,000 L production vessel (SOP-1001). The model and the commercial vessel are both stirred tank bioreactors with equivalent control of pH, dissolved oxygen, temperature and nutrient addition. The scale independent variables of the model, which include the five parameters characterized here, were operated at the values proposed for commercial operation. Scale dependent variables such as agitation, gas flow and headspace pressure were set so that the model matches the commercial vessel on process performance instead of on geometry.

The model qualification compared the model against at-scale data on the input and output attributes that matter to this step: growth trajectory, titre, viability and the released quality attributes. The warrant that qualification gives is specific. Conclusions about parameter ranges and about the position of the operating point relative to the edges of failure transfer from the model to the commercial vessel, because the model reproduces the response of those attributes to the parameters being varied over the ranges studied here. The model is not evidence about commercial scale consistency, which only commercial scale batches can provide.

Reproducibility of the model itself is quantified by the centre point replicates of each design and is reported below (§5.1). Those replicates also supply the pure error term against which lack of fit is tested, which ties the adequacy of every model in §5.3 to the reproducibility of the scale down system.

3.2 Cell line, media and culture operation

The A-Mab production cell line was thawed and expanded through the seed train, and the production bioreactor was inoculated at the initial viable cell concentration given in Table 4.1. Media and feeds were prepared (SOP-2004) from a single lot of each component for the whole campaign, which removes lot to lot medium variation from the measured effects. Basal medium concentration and nutrient feed-1 volume were varied only in the univariate assessment.

The culture was operated as a fed batch (SOP-2003), with pH, temperature and dissolved oxygen under closed loop control and dissolved carbon dioxide controlled through the gas sparge. The culture received nutrient feed-1 on the scheduled profile as a percentage of working volume. Harvest of the culture followed the scheduled duration for the run, and the harvest material was clarified before analysis.

Raw materials controlled to platform specification were treated as identity and quality controlled inputs, not as characterization factors. Control of those materials is a release matter under SOP-2004, and no range for them is claimed by this study.

3.3 Analytical methods

Table 3.1 lists the controlled procedures and validated methods used, of which released N-glycan mapping by 2-AB HILIC-UPLC (SOP-3010, AMV-3010) reports afucosylation, galactosylation and high mannose from one separation. Size variants including aggregate (HMW) are measured by SEC-HPLC (SOP-3011, AMV-3011), charge variants by icIEF (SOP-3013, AMV-3013), host cell protein by ELISA (SOP-3012, AMV-3012) and residual DNA by qPCR (SOP-3014, AMV-3014).

Table 3.1: Controlled procedures and validated analytical methods.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2003	Operation of the Production Bioreactor (Fed-Batch)	SOP
SOP-2004	Cell-Culture Media and Feed Preparation	SOP
SOP-3010	Released N-Glycan Mapping by 2-AB HILIC-UPLC	SOP
SOP-3011	Size-Variant Analysis by SEC-HPLC	SOP
SOP-3012	Host-Cell-Protein Quantitation by ELISA	SOP
SOP-3013	Charge-Variant Analysis by icIEF	SOP
SOP-3014	Residual Host-Cell DNA by qPCR	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3010	N-Glycan Map (2-AB HILIC-UPLC)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation

Method performance sets a floor on the effects this study can resolve. The N-glycan method is validated to an intermediate precision of 2.5 % relative standard deviation with a limit of quantitation of 0.5 % glycan, and the host cell protein ELISA to 9.5 % with a limit of quantitation of 1 ng/mg. An effect smaller than the analytical variability of the assay is reported here as an estimate, not as a demonstrated change. Full

performance of each assay is given in Appendix D and in the validation reports named above.

3.4 Sampling plan

Each run was sampled at harvest, and the harvest pool of every run supplied the material for all 5 designed responses. Sampling and analysis of every run followed the same sequence, and the analytical order was randomized against the run order so that assay drift could not align with a factor setting.

The screening design carries 3 centre point replicates and the response surface design carries 4. Replication at the centre, and not at the corners, is what makes the pure error term independent of the model form, which is why the lack of fit tests in §5.3 have any meaning. Both sets of centre points are separate cultures run to the same conditions, not repeat injections of one harvest.

3.5 Statistical methods

Factors were coded so that the set-point maps to zero and the edges of the characterization range map to minus one and plus one. Coding the factors makes the estimates comparable across parameters with different units, and every effect and coefficient in this report is reported on the coded scale. Analysis followed the statistical procedure for designed experiments (SOP-1002).

The screening design was analysed as a two factor interaction model fitted by ordinary least squares. An effect is reported as twice the coded coefficient, which is the change in the response between the low and the high edge of the characterization range. The response surface design was analysed as a full quadratic model in the 4 retained factors. Terms of both models are judged at $\alpha = 0.05$.

Model adequacy is assessed on four things: R^2 , adjusted R^2 , predicted R^2 from the predicted residual error sum of squares (PRESS), and an analysis of variance that partitions the residual into lack of fit and pure error. A predicted R^2 well below the adjusted value indicates a model that fits the runs it was fitted to and predicts new ones poorly. A lack of fit test on the same model, where it reaches significance, indicates a model form that misses structure the replicates say is real.

Proven acceptable ranges (PAR) were computed in two analyses for each governed attribute and each response surface factor. The first analysis holds the other factors at their set-points and scans the parameter of interest across its characterization range using the fitted response surface model. The second propagates normal operating range variation: the other factors are drawn inside their normal operating ranges, the fitted model is evaluated with the residual variance added, and the range is the region in which the upper bound of the 95 % predictive interval stays within acceptance. The second analysis uses a grid of 81 points with 2,000 draws at each point, and that analysis is the reproducible default reported below (§7).

Commercial scale capability was estimated by Monte-Carlo simulation of 2,000 batches with every parameter drawn inside its normal operating range (NOR). The capability index C_{pk} is reported against the binding limit, which for a one sided criterion is the single limit and for a two sided criterion is the nearer of the two.

4 Study design

4.1 Factors, ranges and the knowledge space

Taken together, the two designed experiments resolve which parameters move the attributes and then model the region those parameters define. Every parameter of the step appears in Table 4.1 with the set-point, the normal operating range, the characterization range, the study type and the final classification. The characterization range is the knowledge space edge and is not itself a proven acceptable range, which is a computed quantity reported per attribute in §7.

Table 4.1: Process parameters, ranges, study type and final classification.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Culture pH	pH	6.85	6.75–6.95	6.6–7.1	WC-CPP	multivariate
Culture temperature	°C	35	34.5–35.5	34–36	WC-CPP	multivariate
Dissolved CO2 (pCO2)	mmHg	100	70–130	40–160	WC-CPP	multivariate
Osmolality	mOsm	400	375–425	360–440	WC-CPP	multivariate
Culture duration	days	17	16–18	15–19	WC-CPP	multivariate
Dissolved oxygen	% sat	50	40–60	30–70	KPP	univariate
Initial viable cell conc.	e6 cells/mL	1	0.8–1.2	0.5–1.5	KPP	univariate
Basal medium concentration	X	1.2	1–1.4	0.8–1.6	GPP	univariate
Nutrient feed-1 volume	% WV	12	10–14	9–15	KPP	univariate

Each characterization range was justified against prior knowledge before execution. The culture pH range of 6.6 to 7.1 spans the settings over which the platform has been operated across three products, and reaches the point at which cell specific productivity begins to fall. The culture temperature range of 34 to 36 °C is wide enough to change the growth and production balance without inducing the cold shock response. The dissolved carbon dioxide range of 40 to 160 mmHg was studied because

accumulation to the upper end of that range is a known consequence of scale and is the mechanism by which large vessels differ from small ones. The osmolality range of 360 to 440 mOsm and the culture duration range of 15 to 19 days complete the set, the latter reaching one full feed cycle either side of the set-point.

Table 4.2: Coded factor legend used in the effect and coefficient tables.

Code	Factor	Unit	Studied in
A	Culture pH	pH	screening + RSM
B	Culture temperature	°C	screening + RSM
C	Dissolved CO ₂ (pCO ₂)	mmHg	screening + RSM
D	Osmolality	mOsm	screening
E	Culture duration	days	screening + RSM

Table 4.2 gives the letter codes used throughout §5 and Appendix C, in screening order, and the same letter denotes the same factor in the response surface model.

4.2 Screening design

The screening design is a two level factorial in the 5 multivariate parameters, augmented with 3 centre points, for 19 runs in all. It estimates all 5 main effects and all two factor interactions between them, and the purpose of a design of that kind is to identify the parameters with the largest statistical effect on the quality attributes. The design is close to saturated at that resolution, and leaves 3 residual degrees of freedom. Estimates from that design are reported as effect sizes and are not used for prediction. The full coded matrix is in an appendix (Appendix A).

4.3 Response-surface design

The response surface design is a face centred central composite design in the 4 factors screening retained, augmented with 4 centre points, for 28 runs. Face centred axial points keep every setting of the design inside the characterization ranges already justified above, which matters here because the ranges were chosen to reach the edges of failure and extending beyond them would have no operational meaning. The design adds curvature to the effects and interactions the screening design resolved, and the resulting full quadratic model is the predictive model used for the design space (§6), the proven acceptable ranges (§7) and the capability estimate (§8). The full coded matrix is given in an appendix (Appendix B).

Because the response surface design does not carry osmolality, the interactions involving that factor are represented in this report by the screening estimates alone. §5.4 states what that costs and §11 carries it forward as a limitation.

4.4 Univariate assessment

Each of dissolved oxygen, initial viable cell conc., basal medium concentration and nutrient feed-1 volume was assessed one parameter at a time across the characterization ranges in Table 4.1, with the remaining parameters held at their set-points. The univariate assessment of those parameters covered the growth and productivity of the culture and the released quality attributes. None of the four parameters was carried into the multivariate design, for the reason RA-001 gives: all four act through the growth trajectory of the culture, and the parameters that act on the chemistry of glycosylation and deamidation are the five in the screening design.

That choice has a boundary, and the boundary is stated here, not in the discussion. Interactions between these four parameters and the five multivariate factors were not studied. The classifications of those parameters in §9 rest on the univariate result and on the platform record, not on a multivariate demonstration, and any future move that changes two of them together sits outside the evidence this report provides.

5 Results

5.1 Centre-point performance and reproducibility

Before any effect is reported, the centre point replicates fix what the study can resolve. Those replicates are summarized in Table 5.1, with the mean, the standard deviation and the relative standard deviation of each response across the 4 centre points of the response surface design.

Table 5.1: Centre-point reproducibility of the response-surface design.

Response	n	Mean	SD	%CV
Afucosylation	4	7.61	0.355	4.66
Galactosylation	4	25.9	2.34	9.02
High mannose	4	5.66	0.559	9.88
Acidic variants	4	21.5	0.877	4.08
Aggregates (HMW)	4	1.51	0.0985	6.53

Reproducibility across those replicates runs from 4.1 % to 9.9 % relative standard deviation across the five responses, and the least precise of them is high mannose against the most precise at acidic variants. The two glycan responses that sit at the high end of that spread are measured on the same separation, whose validated intermediate precision is 2.5 % (AMV-3010). Most of the observed centre point scatter is analytical in origin.

The screening design shows the same picture on 3 replicates, with a highest relative standard deviation of 11.5 % on high mannose. Both replicate sets are consistent with the scale down model reproducing the set-point condition, and both are used as pure error in the analyses that follow.

5.2 Screening: factor effects

Screening resolved a small number of large effects and a larger number of terms that cannot be separated from noise. Table 5.2 gives the six largest effects on galactosylation, which is the response with the most structure.

Table 5.2: Largest screening effects on galactosylation (effect = $2 \times$ coded coefficient).

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
E	-7.82	-3.91	0.193	-20.2	0.000264	***
AE	6.9	3.45	0.193	17.9	0.000382	***
C	-5.79	-2.9	0.193	-15	0.000644	***
A	-4.56	-2.28	0.193	-11.8	0.00131	**
BD	3.7	1.85	0.193	9.57	0.00242	**
AC	3.6	1.8	0.193	9.32	0.00261	**

Galactosylation moves further than any other response across the characterized ranges: culture duration lowers it by 7.8 percentage points from the low edge of the range to the high edge, dissolved carbon dioxide lowers it by 5.8 points, and culture pH by 4.6 points. Culture temperature acts in the opposite direction and raises it by 3.5 points. The interaction between culture pH and culture duration (Table 5.2) is almost as large as the duration main effect, at 6.9 points, which is the first sign that a one dimensional range would misdescribe this response. In all, 12 of the 15 estimated terms reach significance at $\alpha = 0.05$.

Acidic variants are governed by one factor, and dissolved carbon dioxide lowers them by 8.2 percentage points across the characterized range ($p = 0.0004$), which is the largest single effect anywhere in the screening set. Culture pH and culture temperature follow at 4.2 and 3.0 points. However, the attribute is of very low criticality and the acceptance criterion applies over a wide band. The highest acidic variant level any screening run reached was 29.6 %, against a drug substance criterion of 18 to 40 %.

Table 5.3: Largest screening effects on high mannose.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	1.51	0.756	0.155	4.89	0.0164	*
B	-1.39	-0.696	0.155	-4.5	0.0205	*
C	-0.446	-0.223	0.155	-1.44	0.245	
BC	0.369	0.185	0.155	1.19	0.318	
E	0.158	0.0791	0.155	0.511	0.645	
D	-0.139	-0.0693	0.155	-0.448	0.685	

High mannose depends on fewer factors than any other response, and only 2 terms reach significance at $\alpha = 0.05$, both of them main effects (Table 5.3): culture pH raises high mannose by 1.5 percentage points across the characterized range and culture temperature lowers it by 1.4 points. No interaction involving high mannose is separable from noise on this design. The three inactive factors were retained in the knowledge space, because an attribute of very high criticality that does not respond to dissolved CO₂ (pCO₂), osmolality and culture duration over ranges this wide is robust to them, and that robustness is part of the control strategy argument in §10.

Afucosylation and aggregate complete the set, and both carry fewer active factors than galactosylation. Afucosylation is driven by culture duration and by the interaction between culture duration and culture pH, at 2.4 and 2.5 percentage points, which is consistent with a competition that resolves as transit slows. Aggregate is driven mainly by culture pH and culture duration, at 0.83 and 0.58 percentage points, both of which raise it. Neither response shows a significant effect of osmolality.

However, two limits on the screening analysis need stating. The overall model does not reach significance for high mannose ($p = 0.18$), which follows from the 3 residual degrees of freedom the design leaves, not from an absence of structure. Predicted R^2 is negative for every response, with a highest value of -0.44. A near saturated design produces exactly that, because almost every degree of freedom has gone into estimating a term. Screening therefore ranks the factors, and the response surface model estimates the magnitudes.

The 4 factors carried forward were culture pH, culture temperature, culture duration and dissolved CO₂ (pCO₂). Osmolality was dropped despite a resolved main effect on galactosylation and on acidic variants, because those effects are the smallest of the five and because the normal operating range is held to 375 to 425 mOsm by the medium and feed recipe. The full effect tables for all 5 responses are given in an appendix (Appendix C).

5.3 Response-surface models

On the criteria set out in §3.5, the response surface models are adequate for all 5 responses and predictive for four of them, and Table 5.4 gives the summary statistics.

Table 5.4: Response-surface model adequacy (full quadratic, $N = 28$).

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
Afucosylation	28	0.963	0.923	0.795	24.3	5.05e-07	0.584
Galactosylation	28	0.971	0.94	0.915	31.3	1.06e-07	1.36
High mannose	28	0.911	0.815	0.65	9.48	0.000118	0.472
Acidic variants	28	0.938	0.872	0.598	14.1	1.27e-05	1.57
Aggregates (HMW)	28	0.969	0.936	0.835	29.3	1.58e-07	0.0985

Across the five responses, R^2 runs from 91.1% to 97.1% and adjusted R^2 does not fall below 0.81. As a result, the fitted terms are carrying real structure and not degrees of freedom. Predicted R^2 is the more demanding statistic, and that statistic separates the five responses. Four of them reach 0.65 or better, with the strongest at 0.92 for galactosylation, and the weakest of the five is acidic variants at 0.60. A model at that level predicts the mean response usefully and predicts an individual batch loosely, and §7 treats it accordingly.

Table 5.5: Analysis of variance for the galactosylation response-surface model.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	811	14	57.9	31.3	1.06e-07
Residual	24.1	13	1.85	nan	nan
Lack of fit	7.69	10	0.769	0.141	0.992
Pure error	16.4	3	5.46	nan	nan

Table 5.5 partitions the galactosylation model, in which the model term carries 97.1% of the total sum of squares, and the residual splits into 10 degrees of freedom of lack of fit against 3 of pure error. The lack of fit test does not approach significance ($p = 0.99$). In addition, every other response behaves the same way, and the lowest lack of fit probability of the five stays well above $\alpha = 0.05$.

That test has a bound worth stating. The pure error term rests on 3 degrees of freedom from 4 centre point replicates. Therefore a non significant result bounds the evidence for the model form without establishing it. For this reason the weakest case is acidic variants ($p = 0.15$), and it is the one to watch when the models are challenged at commercial scale.

Table 5.6: Full quadratic coefficients for galactosylation (coded factors).

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	25.8	0.47	54.8	9.11e-17	***
A	-2.21	0.321	-6.89	1.09e-05	***
B	2.3	0.321	7.18	7.13e-06	***
E	-3.82	0.321	-11.9	2.27e-08	***
C	-2.43	0.321	-7.56	4.11e-06	***
AB	1.67	0.34	4.92	0.000279	***
AE	2.72	0.34	7.99	2.28e-06	***
AC	2.35	0.34	6.9	1.09e-05	***
BE	-0.217	0.34	-0.639	0.534	
BC	0.315	0.34	0.928	0.37	
EC	0.332	0.34	0.975	0.347	
A ²	-0.837	0.847	-0.988	0.341	
B ²	0.378	0.847	0.446	0.663	
E ²	0.937	0.847	1.11	0.289	
C ²	-0.373	0.847	-0.441	0.667	

Table 5.6 gives the galactosylation coefficients in model order, of which 7 of the 14 terms reach significance. The four main effects hold the signs the screening design gave them, and three interactions survive: culture pH with culture duration (2.72), culture pH with dissolved carbon dioxide (2.35) and culture pH with culture temperature (1.67). No quadratic term reaches significance. Consequently, the galactosylation surface is a plane twisted by the interactions, not a curved one.

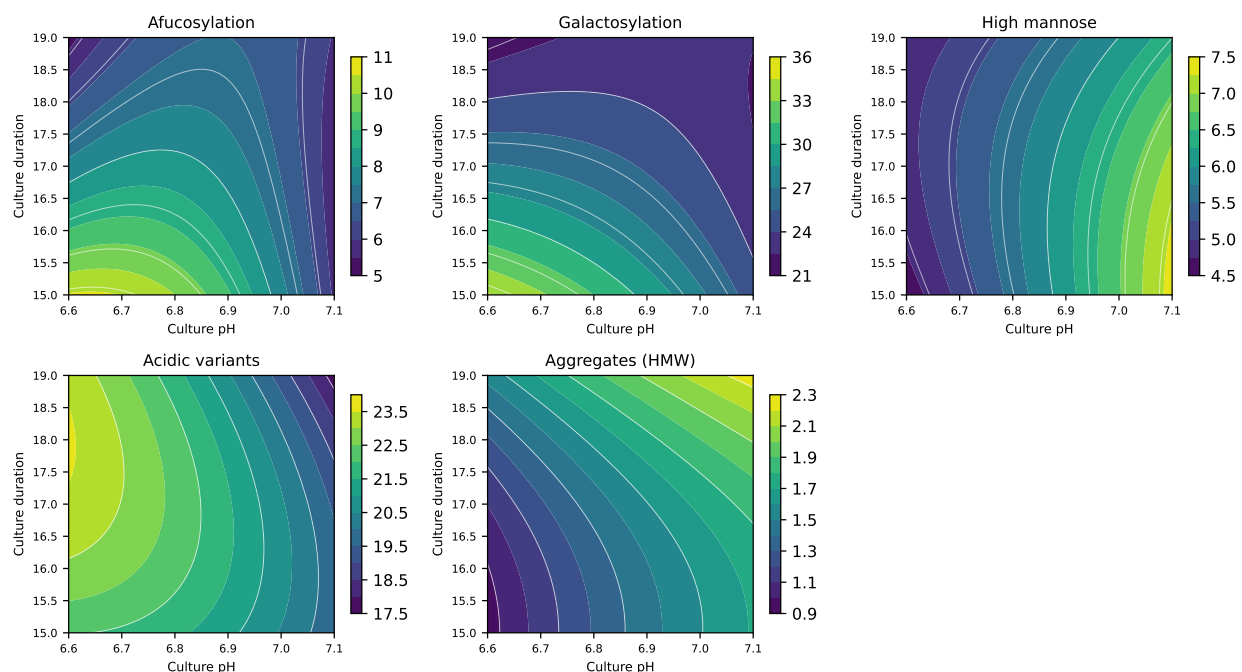
Table 5.7: Full quadratic coefficients for high mannose (coded factors).

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	5.89	0.163	36	2.06e-14	***
A	1.09	0.111	9.77	2.36e-07	***
B	-0.421	0.111	-3.78	0.00227	**
E	-0.188	0.111	-1.69	0.115	
C	0.212	0.111	1.91	0.0788	
AB	0.148	0.118	1.26	0.231	
AE	-0.29	0.118	-2.46	0.0289	*
AC	0.293	0.118	2.48	0.0276	*
BE	0.0874	0.118	0.74	0.472	
BC	-0.118	0.118	-0.996	0.337	
EC	-0.0402	0.118	-0.341	0.739	
A ²	0.116	0.294	0.393	0.701	
B ²	0.094	0.294	0.32	0.754	
E ²	-0.215	0.294	-0.731	0.478	
C ²	0.126	0.294	0.43	0.674	

High mannose keeps the simpler structure that screening found: culture pH dominates at 1.09 per coded unit, culture temperature follows at -0.42, and two small interactions with culture pH reach significance on the larger design. The additional runs of the response surface design resolve those interactions where the screening design could not, which is the improvement in resolution the second stage was run for.

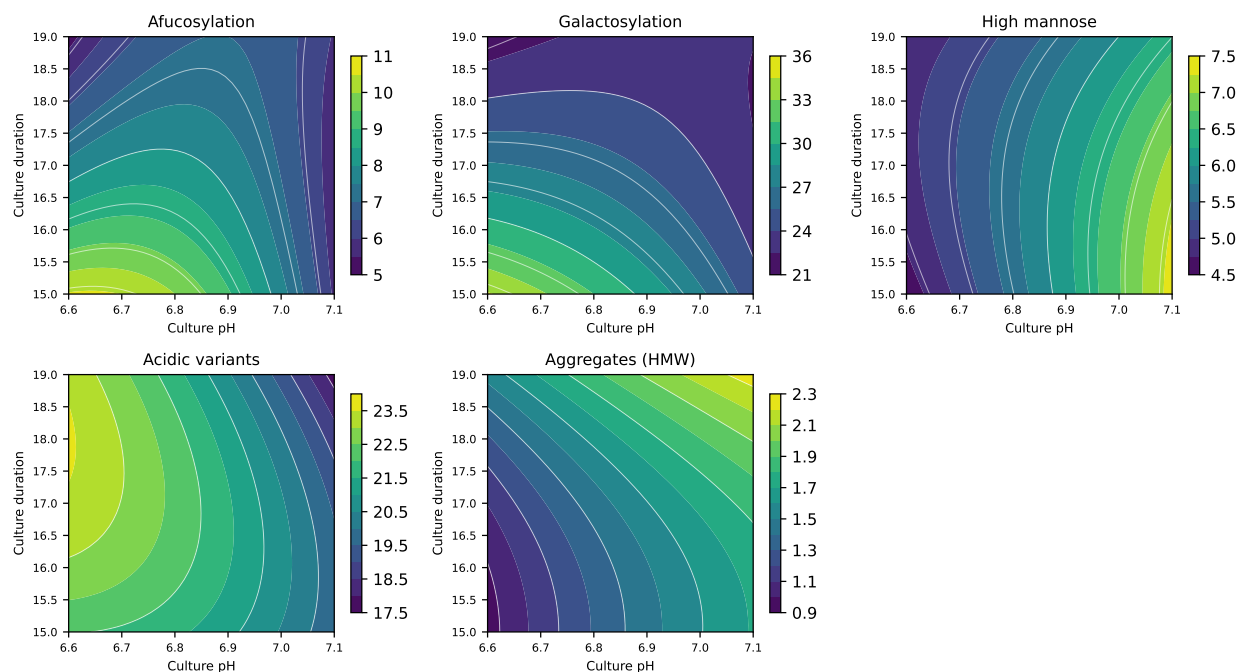
Afucosylation is the one response with a significant quadratic term, and the coefficient on the squared culture pH term is -1.37 ($p = 0.002$). Afucosylation therefore falls away at both ends of the pH range instead of tracking it linearly. Aggregate carries a smaller curvature in culture duration (0.14, $p = 0.037$). Coefficient tables for the remaining responses are given in an appendix (Appendix C).

Response-surface predictions (remaining factors held at set-point)



(a) Response-surface predictions over culture pH and culture duration, with culture temperature and dissolved CO₂ at their set-points.

Response-surface predictions (remaining factors held at set-point)

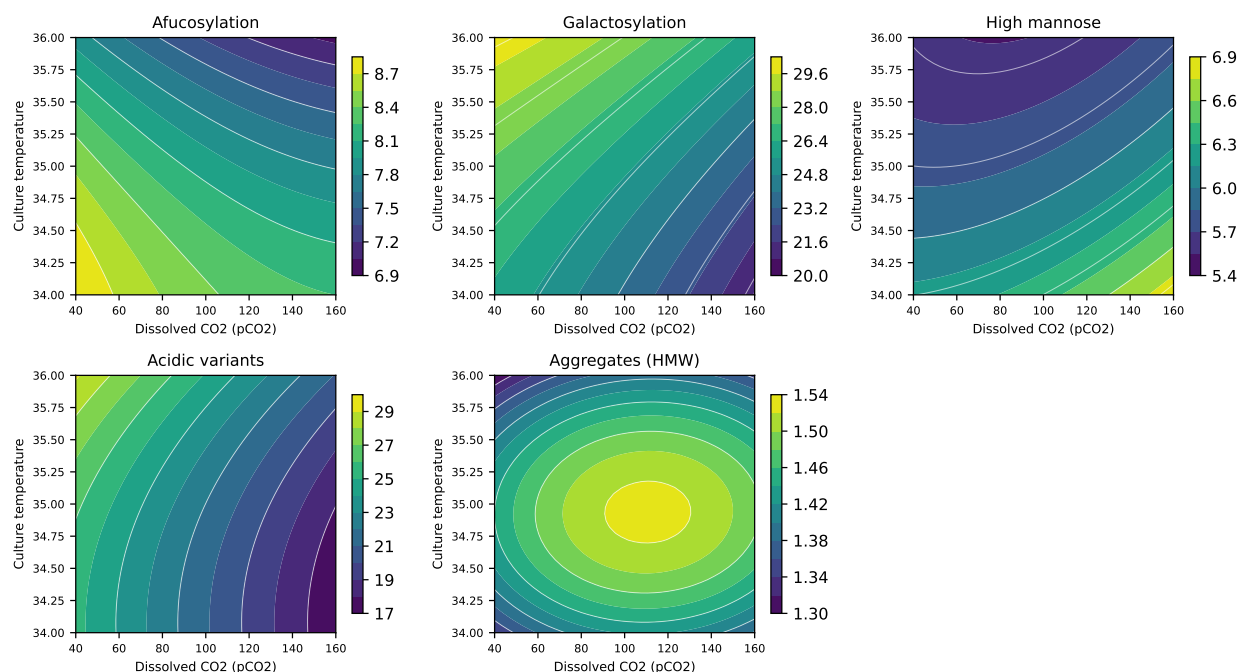


(b)

Figure 5.1

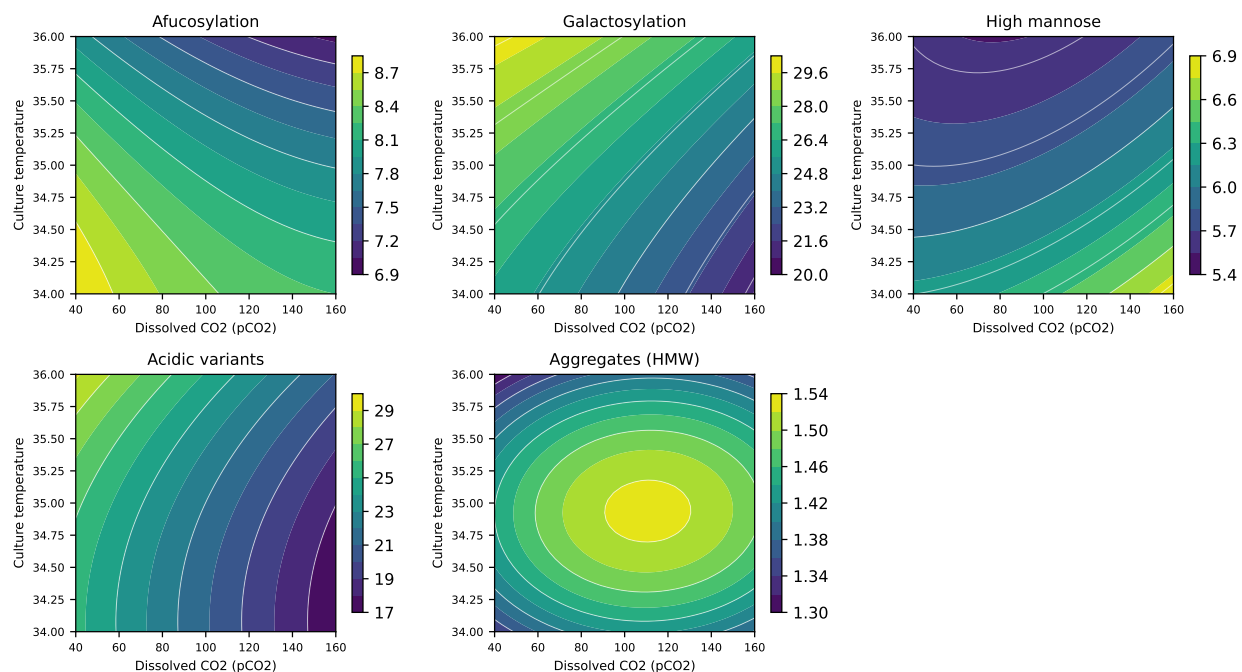
Figure 5.1 shows all 5 surfaces over culture pH and culture duration, the two factors that appear in the most models. The galactosylation and afucosylation panels show the twist that the pH by duration interaction produces: the pH gradient at the short duration edge runs the opposite way from the gradient at the long edge. High mannose and aggregate show simple gradients in pH, and acidic variants varies with pH but hardly at all with culture duration.

Response-surface predictions (remaining factors held at set-point)



(a) Response-surface predictions over dissolved CO₂ and culture temperature, with culture pH and culture duration at their set-points.

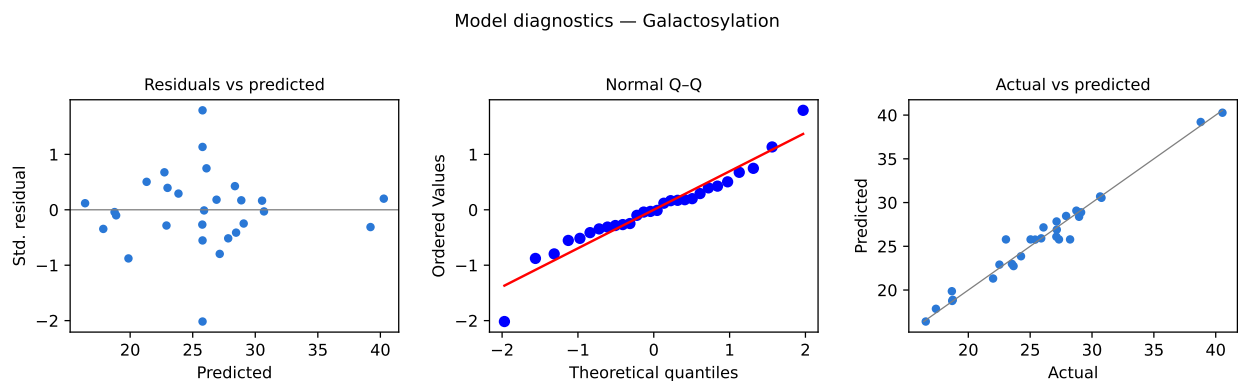
Response-surface predictions (remaining factors held at set-point)



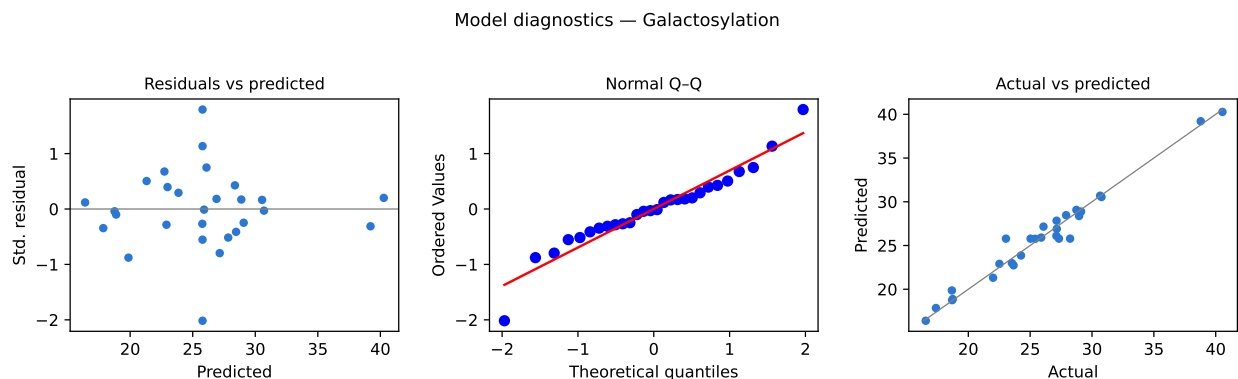
(b)

Figure 5.2

Figure 5.2 covers the other two factors, of which acidic variants is the response those two govern, and the panel shows the strong dissolved carbon dioxide gradient with the weaker temperature gradient across it. Galactosylation also falls steadily with dissolved carbon dioxide, which makes that parameter the one factor that moves a high criticality glycan and the low criticality charge variant together.

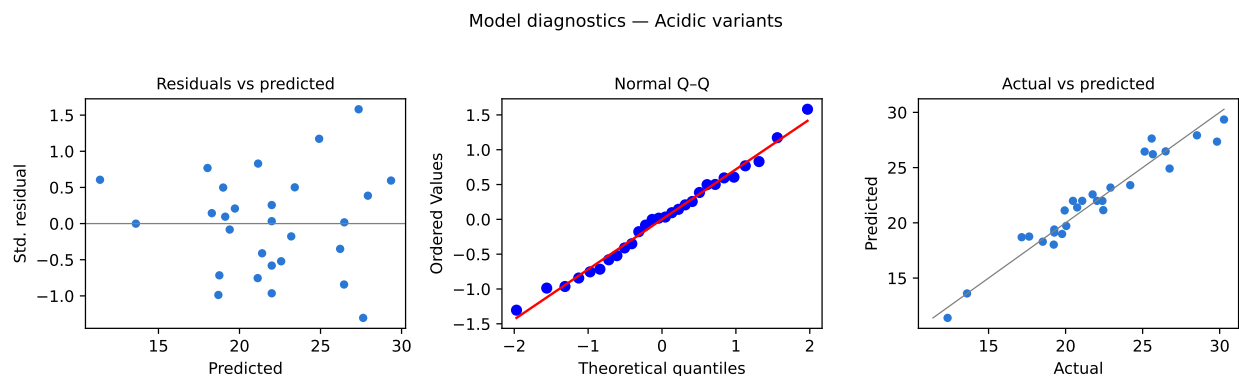


(a) Model diagnostics for galactosylation: residuals against predicted values, normal quantile plot and actual against predicted.

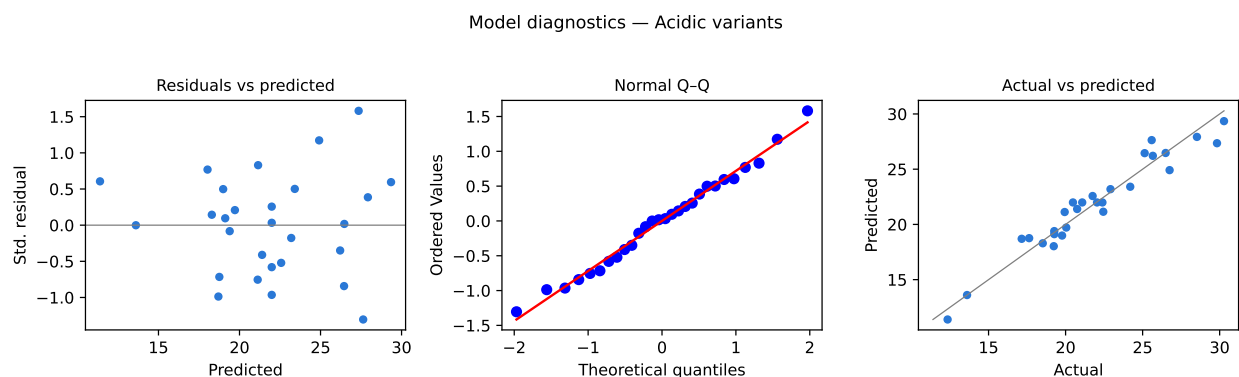


(b)

Figure 5.3



(a) Model diagnostics for acidic variants, the response with the lowest predicted R^2 .



(b)

Figure 5.4

Figure 5.3 and Figure 5.4 give the diagnostics for the strongest and the weakest of the five models. In both, the residuals scatter about zero with no trend against the predicted value and no fan, and the actual against predicted points sit on the identity line. The largest standardized residual in the galactosylation model is 2.0 and in the acidic variants model 1.6, and neither residual set departs from normality (Shapiro-Wilk $p = 0.19$ and 0.97). No run was excluded from any fit.

5.4 Mechanistic interpretation

The surfaces behave as the platform mechanisms predict, with one exception that the study quantifies without explaining. Galactosylation falls as the culture ages, which is what a declining galactosyltransferase activity and a depleting nucleotide sugar pool produce. Dissolved carbon dioxide lowers it further, and the accepted mechanism for that is an acidified Golgi lumen: transferase activity and nucleotide sugar transport both fall as luminal pH drops away from the enzyme optimum. Culture temperature raises galactosylation in the same picture, through the rate constant of the transfer reaction.

The pH by duration interaction on galactosylation follows from the same account. At the short duration edge of the range the transferase is still active, and raising culture pH lowers the product by 9.9 percentage points. At the long duration edge the same move changes it by 1.0 points. Once the enzyme system is depleted, the pH of the culture has little left to act on. An operating region built from one dimensional ranges would not represent that, which is the argument for the multivariate region (§6).

High mannose behaves as an incomplete trimming attribute: the Golgi mannosidases work best at an acidic luminal pH, and raising the culture pH raises the luminal pH away from that optimum, which leaves more of the untrimmed precursor in the released product. Higher culture temperature drives trimming further towards completion and lowers high mannose. Neither dissolved carbon dioxide nor culture duration moves the attribute measurably, which is consistent with trimming reaching the endpoint well before harvest.

Afucosylation reflects a competition between core fucosyltransferase activity and the rate at which antibody transits the secretory pathway. Longer culture lowers afucosylation, because slower transit gives the transferase more time to act on each molecule. The curvature in culture pH says that the competition has an optimum inside the range instead of at an edge: the highest predicted afucosylation over a pH scan at the set-point of the other factors is 8.2 %, and the level falls at both edges.

Aggregate is not enzymatic. Antibody that sits longer in an ageing broth, at a higher pH and in the presence of proteases released by dying cells, aggregates more, and both significant factors act in that direction. The effect is small in absolute terms across the whole characterized range, at 0.83 percentage points for culture pH. Aggregate is nonetheless worth close attention here, as the only attribute of the five that downstream steps also change, and the cation exchange step is sized against the level this step delivers (PCR-007).

Acidic variants carry the exception. Dissolved carbon dioxide lowers them, and the deamidation mechanism accounts for that directly, since an acidified culture fluid slows a base catalysed reaction, and culture temperature raises them in the way the rate constant predicts. However, culture pH lowers acidic variants as well, and solution chemistry predicts the opposite. The most likely account is an indirect one through cell physiology and through the age distribution of product in the harvest, and the study does not separate those possibilities. The response surface model quantifies the effect without attributing it, and the classification of culture pH in §9 rests on the effects it has on the glycan attributes.

Finally, dropping osmolality from the response surface design has a cost that belongs here. Screening resolved an interaction between culture temperature and osmolality on galactosylation of 3.7 percentage points ($p = 0.002$). That term is represented in this report by the screening estimate alone, and no quadratic or predictive statement is made about that term. The narrow recipe control of osmolality is what keeps the omission acceptable, and §11 carries it as a limitation.

6 Design space

The design space of the production bioreactor is a multivariate region in culture pH, culture temperature, culture duration and dissolved CO₂ (pCO₂), over which the fitted response surface models predict that every attribute the step governs stays within the in-process criterion. The region is a subset of the characterized knowledge space in Table 4.1, and the normal operating ranges sit inside the region over almost all of their extent (CMC Biotech Working Group 2009). One corner of the normal operating range does not, and §7 reports it.

The criteria the region is judged against are the step's in-process limits, not the drug substance specifications. For 4 of the 5 responses the in-process limit is an alert limit set at 2.5 standard deviations above the simulated commercial scale mean, because those attributes are neither cleared nor modified downstream and the drug substance criterion sits far outside the spread of the process itself. For aggregate the limit is carried back from the drug substance criterion through the clearance the downstream train delivers, divided by an assurance margin of 6.8. Both bases give a limit that binds well inside the drug substance criterion, and the in-process limits used in §7 and here are the more demanding comparison of the two available.

Evaluated on an evenly spaced grid of 14,641 points across the four characterized ranges, 57 % of the region meets every criterion simultaneously. The largest share of the grid is rejected by galactosylation, at 14 % of the points, and the smallest by high mannose, at 5 %. The characterized ranges therefore contain edges of failure, which is the intended outcome of choosing ranges wider than the control ranges.

The worst case corner of the characterized region is the one that fails galactosylation: culture pH 6.6, culture temperature 36 °C, culture duration 15 days, dissolved CO₂ (pCO₂) 40 mmHg. At that corner the model predicts 40.3 %, against an in-process limit of 30.1 % and a drug substance criterion of 10 to 40 %. That single corner exceeds even the drug substance criterion, which is what an edge of failure means and why the design space is smaller than the knowledge space.

The principal planes of the region are culture pH against culture duration, shown in Figure 5.1, and dissolved carbon dioxide against culture temperature, shown in Figure 5.2. The first plane carries the interaction that drives galactosylation and afucosylation, and the second carries the gradient that drives acidic variants. A design space stated as four independent ranges would miss the first of those entirely, because the acceptable pH range at the short duration edge is not the acceptable pH range at the long one.

Under ICH Q8, operation anywhere inside the region is not a change from a regulatory filing perspective (International Council for Harmonisation 2009). Movement within the region still runs through the pharmaceutical quality system as a planned change

with a prospective risk assessment (International Council for Harmonisation 2008), and movement outside the region requires change control and supporting data. The normal operating ranges in Table 4.1 are where the commercial process is expected to run, and they occupy a small part of the region, with the exception noted in §7.

Three bounds apply to this claim. The region is defined by mean predictions from models fitted over the characterized ranges, and says nothing about settings outside those ranges. The models were fitted on a qualified scale down system, and commercial scale confirmation is a Stage 2 activity. The region covers the four response surface factors only, with osmolality and the four univariately assessed parameters held inside the ranges in Table 4.1.

7 Proven acceptable ranges

For a single parameter at a time, proven acceptable ranges complement the multivariate region with a per attribute statement of how far a single parameter may move. Table 7.1 gives both analyses for every combination of governed attribute and response surface factor: the range proven with the other factors at their set-points, and the range that survives normal operating range co-variation of the other factors.

Table 7.1: Proven acceptable ranges per governed attribute and response-surface factor.

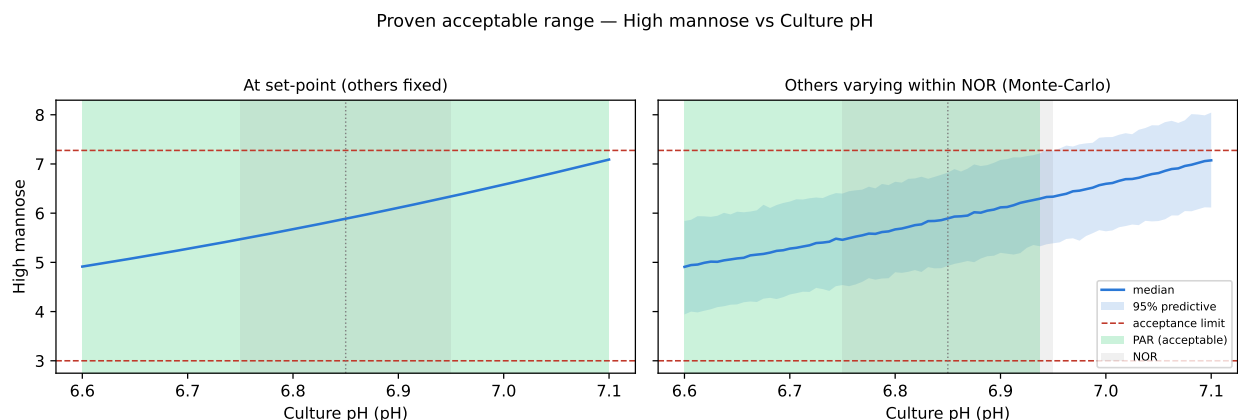
CQA	Parameter	Char. range	Unit	Crite- rion	PAR (set-point)	PAR (NOR)
Afucosylation	Culture pH	6.6-7.1	pH	≤ 9.43	6.6-7.1	6.82-7.1
Afucosylation	Culture temperature	34-36	°C	≤ 9.43	34-36	34.7-36
Afucosylation	Culture duration	15-19	days	≤ 9.43	15.2-19	16.8-19
Afucosylation	Dissolved CO2 (pCO2)	40-160	mmHg	≤ 9.43	40-160	82-160
Galactosylation	Culture pH	6.6-7.1	pH	≤ 30.1	6.6-7.1	6.74-7.1
Galactosylation	Culture temperature	34-36	°C	≤ 30.1	34-36	34-35.5
Galactosylation	Culture duration	15-19	days	≤ 30.1	15.2-19	16.4-19
Galactosylation	Dissolved CO2 (pCO2)	40-160	mmHg	≤ 30.1	40-160	74.5-160
High mannose	Culture pH	6.6-7.1	pH	≤ 7.28	6.6-7.1	6.6-6.94
High mannose	Culture temperature	34-36	°C	≤ 7.28	34-36	34.1-36
High mannose	Culture duration	15-19	days	≤ 7.28	15-19	15-19
High mannose	Dissolved CO2 (pCO2)	40-160	mmHg	≤ 7.28	40-160	40-156
Acidic variants	Culture pH	6.6-7.1	pH	≤ 25.8	6.6-7.1	6.82-7.1
Acidic variants	Culture temperature	34-36	°C	≤ 25.8	34-36	34-35.2

CQA	Parameter	Char. range	Unit	Crite- rion	PAR (set-point)	PAR (NOR)
Acidic variants	Culture duration	15-19	days	≤ 25.8	15-19	15-19
Acidic variants	Dissolved CO2 (pCO2)	40-160	mmHg	≤ 25.8	46.6-160	92.5-160
Aggregates (HMW)	Culture pH	6.6-7.1	pH	≤ 1.86	6.6-7.1	6.6-6.93
Aggregates (HMW)	Culture temperature	34-36	°C	≤ 1.86	34-36	34-36
Aggregates (HMW)	Culture duration	15-19	days	≤ 1.86	15-18.7	15-17.7
Aggregates (HMW)	Dissolved CO2 (pCO2)	40-160	mmHg	≤ 1.86	40-160	40-160

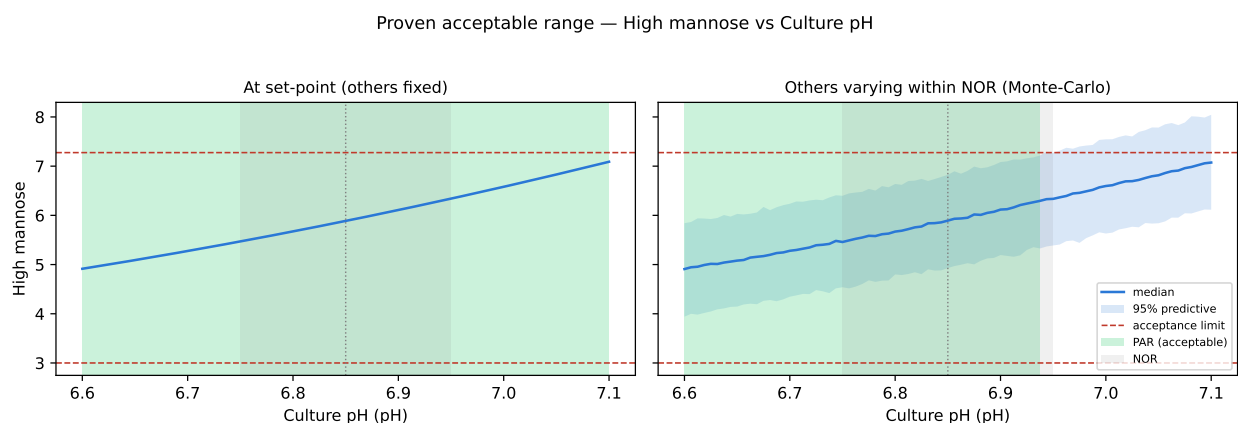
The acceptance basis is stated in the Criterion column and is the same in-process limit used for the design space (§6). Reading the table against the drug substance criteria instead would widen every range in the table.

With the other factors at their set-points, 16 of the 20 combinations return the entire characterization range as a proven acceptable range. The four that do not are informative. Afucosylation and galactosylation both restrict culture duration at the short end, to 15.2-19 days and 15.2-19 days against a characterized range of 15 to 19 days. Acidic variants restrict dissolved carbon dioxide at the low end, to 46.6-160 mmHg against a characterized range of 40 to 160 mmHg. Aggregate restricts culture duration at the long end, to 15-18.7 days.

Under normal operating range co-variation the ranges narrow further, and only 4 of the 20 combinations still span the full characterization range. Narrowing is the expected outcome, because this analysis compares the upper bound of a 95 % predictive interval against the limit instead of a mean prediction, and the interval carries both the co-variation of the other factors and the residual variance of the model.



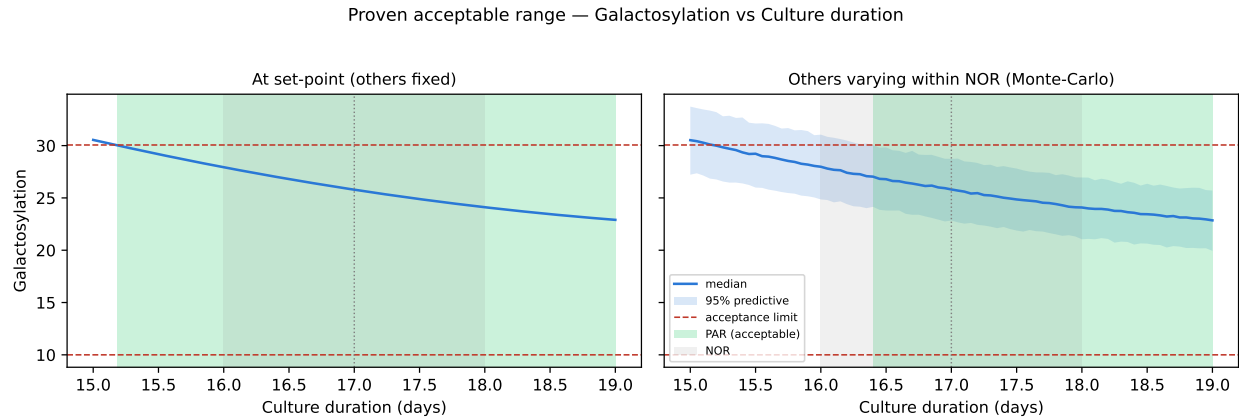
(a) Proven acceptable range for high mannose against culture pH, at set-point (left) and with the other factors varying within their normal operating ranges (right). The acceptable region is shaded green.



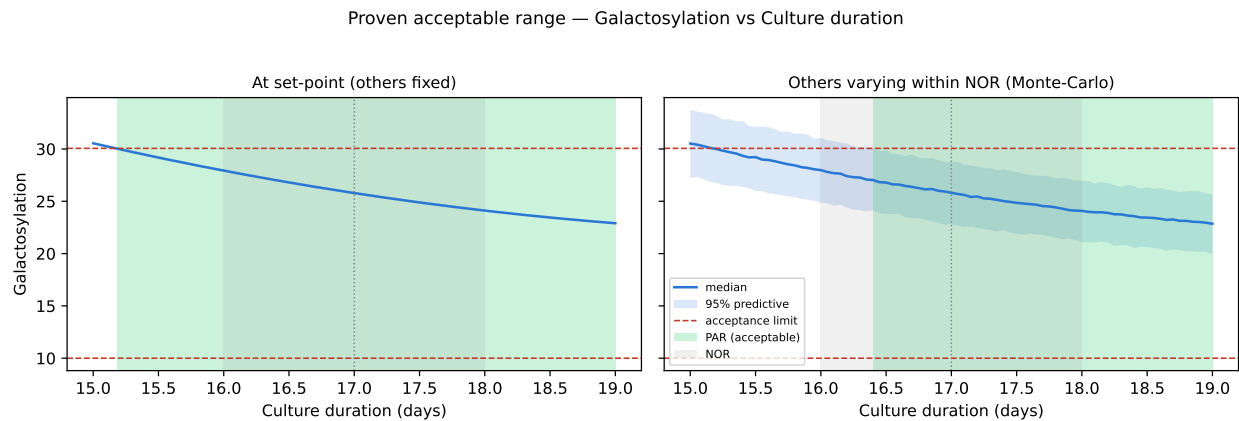
(b)

Figure 7.1

Figure 7.1 gives the analysis for high mannose against culture pH, which is the factor with the largest response surface main effect on it. At set-point the whole characterized pH range is acceptable, and under normal operating range co-variation the acceptable range becomes 6.6–6.94. The upper part of the characterized pH range no longer holds the predictive bound below the in-process limit. The normal operating range for culture pH is 6.75 to 6.95, and the upper edge of that range sits just above the co-varied bound.



(a) Proven acceptable range for galactosylation against culture duration, at set-point and under normal operating range co-variation.



(b)

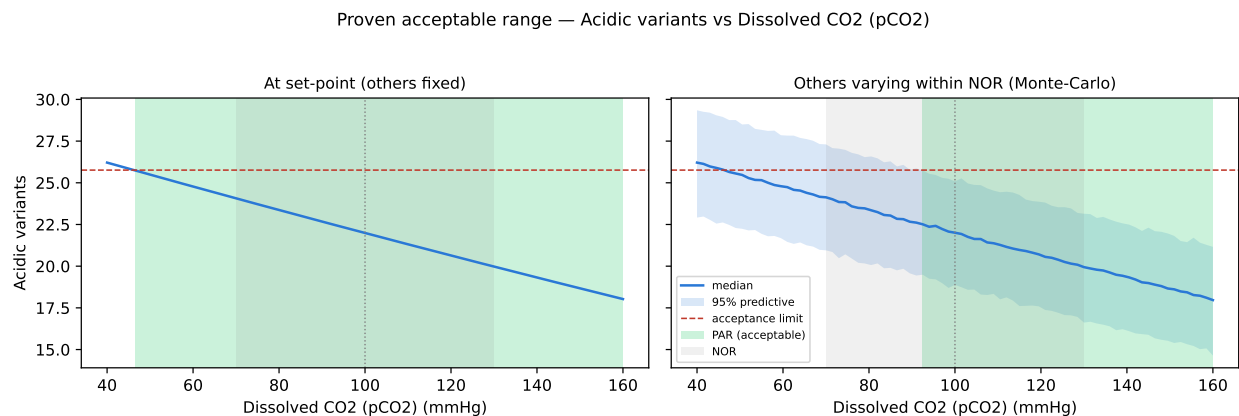
Figure 7.2

Galactosylation is the binding attribute, and Figure 7.2 shows why. Under co-variation the proven acceptable range for culture duration is 16.4–19 days, and the normal operating range for culture duration is 16 to 18 days. The lower edge of the normal operating range therefore sits below the lower edge of the proven acceptable range. The same pattern appears against dissolved carbon dioxide, where the range is 74.5–160 mmHg against a normal operating range of 70 to 130 mmHg. It appears again against culture temperature, where the co-varied range of 34–35.5 °C stops just below the upper edge of the normal operating range of 34.5 to 35.5 °C.

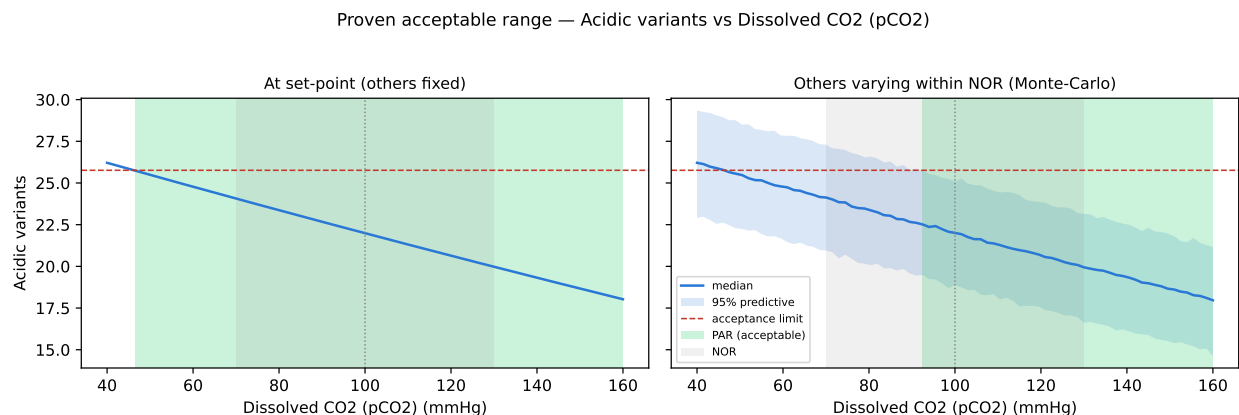
That is an adverse result, quantified here before it is dispositioned. At the corner of the normal operating range where all four factors take their least favourable values at once, culture pH 6.75, culture temperature 35.5 °C, culture duration 16 days, dissolved CO₂ (pCO₂) 70 mmHg, the model predicts galactosylation of 31.8 % against an in-process alert limit of 30.1 %. Afucosylation reaches its own alert limit of 9.4 % at the equivalent corner.

Three things bound that finding. The limit exceeded is an alert limit derived from the demonstrated capability of the process, at 2.5 standard deviations above the simulated

mean, and a limit of that construction is expected to be exceeded by a small proportion of batches. The comparison is made at a simultaneous four factor extreme, whereas the capability simulation draws the parameters independently: over 2,000 simulated batches the highest galactosylation was 32.2 %. Against the drug substance criterion of 10 to 40 % there is no exceedance anywhere in the normal operating range. An excursion of this kind is therefore an investigation trigger under the in-process control strategy in §10, not a specification failure.



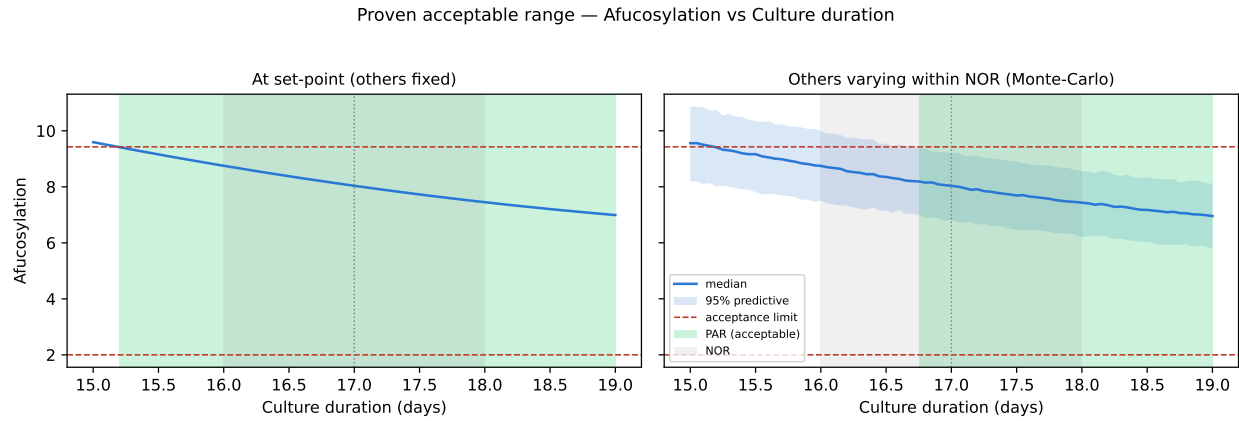
(a) Proven acceptable range for acidic variants against dissolved CO₂, at set-point and under normal operating range co-variation.



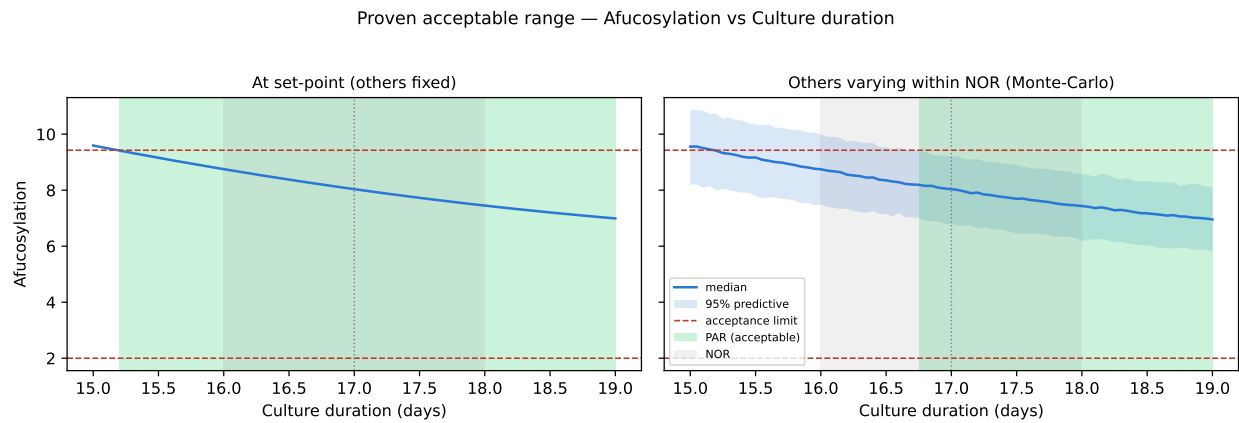
(b)

Figure 7.3

Figure 7.3 gives the acidic variant analysis against dissolved CO₂ (pCO₂), where the at set-point range starts at 46.6–160 mmHg and the co-varied range at 92.5–160 mmHg, both bounded at the low end because low dissolved carbon dioxide raises the attribute. The normal operating range of 70 to 130 mmHg extends below the co-varied bound, which is the same pattern as galactosylation and has the same disposition.



(a) Proven acceptable range for afucosylation against culture duration, at set-point and under normal operating range co-variation.



(b)

Figure 7.4

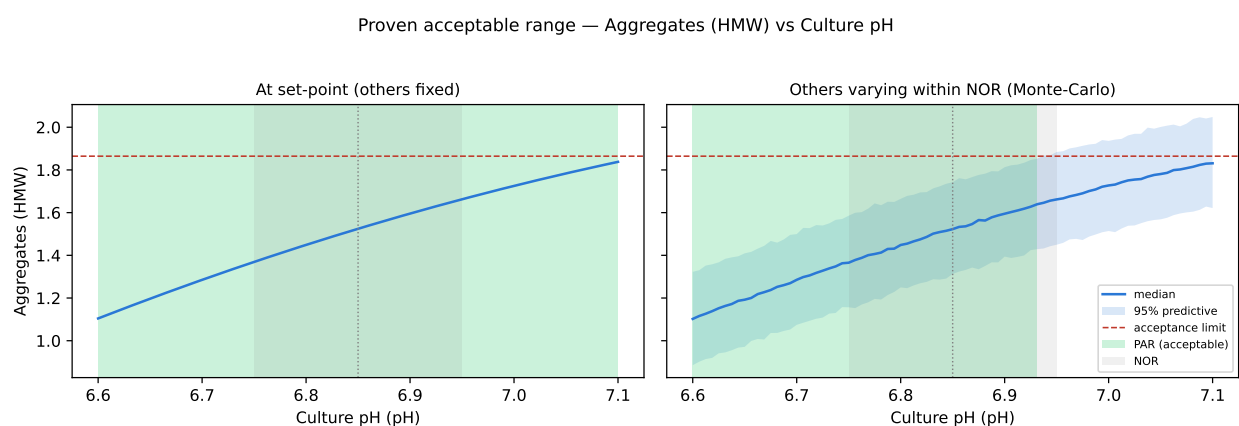
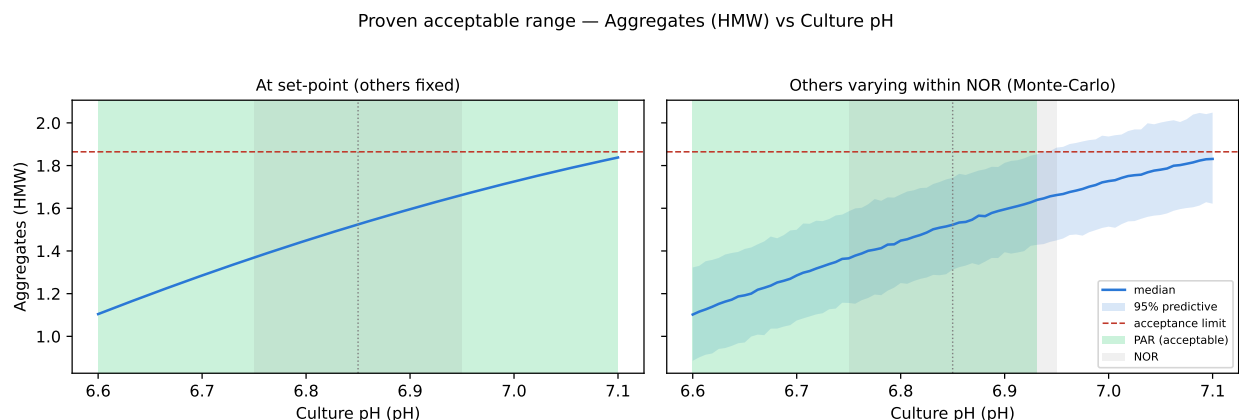


Figure 7.5

Figure 7.4 and Figure 7.5 complete the set, and afucosylation restricts culture duration at the short end under both analyses, at 15.2–19 days and 16.8–19 days. Aggregate restricts culture pH only under co-variation, at 6.6–6.93, and the at set-point range covers the whole characterized span.

The relationship between these ranges and the design space needs stating explicitly. A proven acceptable range is univariate, and each one holds only under the analysis that produced it. The design space in §6 is the multivariate region, and where the whole characterization range of a parameter is proven acceptable, the constraint that actually binds the operating region is the multivariate corner, not any univariate range. Both statements rest on the fitted response surface models and on the characterized ranges, and neither extends beyond them.

8 Process capability and robustness

For every attribute in Table 2.1, capability was estimated by simulating 2,000 commercial scale batches with every parameter drawn independently inside the normal operating range and with the fitted models and the residual variance applied, and Table 8.1 gives the result.

Table 8.1: Estimated commercial-scale capability for the attributes the bioreactor governs.

CQA	Crit.	Spec	Acceptance	Mean $\pm$ SD	Cpk
Afucosylation	H	two_sided	2-13	7.98 ± 0.58	2.89
Galactosylation (total %Gal)	H	two_sided	10-40	26 ± 1.6	2.86
High Mannose	VH	two_sided	3-10	5.99 ± 0.51	1.94
Aggregates (HMW)	H	upper	0-5	0.753 ± 0.088	16.1
Acidic charge variants (deamidation)	VL	upper	18-40	22.3 ± 1.4	4.26
Host Cell Protein (HCP)	M-H	upper	0-100	15.2 ± 4.6	6.14
Residual DNA	VL	upper	0-0.001	0.0001 ± 0	10.8

Across those batches, every attribute clears the drug substance criterion. The tightest capability of the 7 attributes is the one for high mannose, at a Cpk of 1.94. The simulated mean behind that capability, 5.99 %, sits 2.99 percentage points above the lower acceptance limit of 3 %, which is 5.8 standard deviations of the simulated distribution. The lower limit binds, not the upper one, and the highest value in 2,000 simulated batches was 8.0 % (Table 8.1).

The reason high mannose is the tightest case is not that the process controls the attribute poorly. High mannose carries the narrowest acceptance band relative to the level the process runs at, at 3 to 10 % around a mean of 6.0 %, and the attribute also sits at very high criticality. Both facts about the attribute point the control strategy at the two parameters that move it, which is what §10 does.

The widest capability is the one for aggregates (hmw), at a Cpk of 16.1, and an index of that size does not mean the process is 8 times better controlled for aggregate than for high mannose. The index means the attribute sits far from the limit that applies to it, at a simulated mean of 0.75 % against a criterion of 0 to 5 %, and the index should be read as showing that aggregate does not constrain this step.

Host cell protein and residual DNA appear in Table 8.1 as drug substance outcomes, not as bioreactor outputs. The bioreactor sets the burden entering the purification

train, and the capability shown for both is the result of that burden and the clearance the train delivers. The clearance credit belongs to Protein A capture (PCR-005), cation exchange (PCR-007) and anion exchange (PCR-008), and the consolidated position is in PCMR-001.

Two bounds apply to every index in Table 8.1. Each is estimated from a qualified scale down model with the parameters inside their normal operating ranges, and none is a measurement at commercial scale. The simulation also assumes that the parameters vary independently inside those ranges, which is why the corner analysis in §7 is reported alongside it.

9 Parameter classification

Classification follows the decision logic for process parameters (SOP-4001) and the continuum of criticality in (International Council for Harmonisation 2023). A parameter whose variability affects a critical quality attribute is critical, and the distinction between a critical process parameter and a well controlled critical process parameter is the risk of leaving the design space, not the size of the effect (CMC Biotech Working Group 2009). The outcome for every parameter appears in Table 4.1, and the rationale for each is below.

- Culture pH is a WC-CPP because it reaches significance on all 5 responses of the response surface model, including the largest effect on high mannose, and because the control system holds it to 6.75 to 6.95 against a proven acceptable range that covers the whole characterized span at set-point (Table 7.1).
- Culture temperature is a WC-CPP because it moves galactosylation, high mannose and acidic variants, and jacketed control holds it to 34.5 to 35.5 °C inside a characterized range of 34 to 36 °C.
- Dissolved carbon dioxide is a WC-CPP because it carries the largest effect in the study on acidic variants and a substantial effect on galactosylation, and sparge control holds it to 70 to 130 mmHg. Dissolved carbon dioxide is also the parameter most exposed to scale, which is why the characterized range reaches 40 to 160 mmHg.
- Osmolality is a WC-CPP because screening resolved effects on galactosylation and acidic variants, and although the parameter was not carried into the response surface design, the medium and feed recipe holds it to 375 to 425 mOsm. Classification on the screening result and on the recipe control (SOP-2004) is the conservative reading of a factor that was not modelled further.
- Culture duration is a WC-CPP because it carries the largest main effect on galactosylation and on afucosylation and sets the harvest impurity burden of the step, and the harvest schedule holds it to 16 to 18 days inside a characterized range of 15 to 19 days.
- Dissolved oxygen is a KPP because the univariate assessment linked it to growth and productivity with no measurable link to a quality attribute, and the vessel controls it to 40 to 60 % of saturation (process consistency depends on it).
- Initial viable cell concentration is a KPP on the same basis, since the seed train delivers it to 0.8 to 1.2×10^6 cells/mL and the parameter sets the growth trajectory without changing the released profile.
- Nutrient feed-1 volume is a KPP because the feed schedule delivers 10 to 14 % of working volume and the parameter governs titre with no measured effect on product quality.
- Basal medium concentration is a GPP because the univariate assessment over 0.8 to 1.6 X found no link to product quality, and none to process performance

that would require control beyond the recipe (SOP-2004).

No parameter at this step is classified as a critical process parameter. The 5 quality linked parameters are all held by the control system inside ranges this study proved acceptable, the risk of an unnoticed excursion outside the design space is correspondingly low, and that outcome matches the platform record for the step (CMC Biotech Working Group 2009).

Finally, null results were classified rather than dropped. Osmolality, dissolved carbon dioxide and culture duration produce no measurable effect on high mannose across ranges that are 1.6 to 2.5 times the width of the control ranges, and that absence is evidence of robustness for the attribute of highest criticality at this step. Those three culture parameters stay in the knowledge space on that basis, and the evidence is bounded by the ranges screened.

10 Contribution to the control strategy

Within the control strategy for A-Mab drug substance, the production bioreactor contributes four elements, and each is stated below with the range or the document that carries it.

The first element is parameter control: 5 quality linked parameters are controlled to the normal operating ranges in Table 4.1, which lie inside the design space (§6) apart from the corner identified in §7. Two of those parameters, culture pH and culture temperature, carry the most weight, because they are the two that move high mannose, the attribute with the tightest capability at this step.

The second element is in-process monitoring against the limits used in §6 and §7, which sit inside the drug substance criteria by construction, at 2.5 standard deviations above the simulated mean for the attributes the train does not modify. An excursion above one of those limits triggers investigation of the culture rather than rejection of the batch, and the corner analysis in §7 shows which combinations of parameter settings would produce one.

The third element is release testing, under which the glycan profile, the charge variant profile and the size variant profile of the drug substance are tested against the criteria in Table 2.1 by the methods in Table 3.1 (AMV-3010, AMV-3011, AMV-3013). Testing confirms the outcome; the parameter ranges and the in-process limits are what assure it.

The fourth element is the input the step provides to the purification train, where host cell protein and residual DNA leave this step far above the drug substance criteria and culture duration is the parameter that sets both burdens. Holding duration to 16 to 18 days therefore bounds the load that Protein A capture, cation exchange and anion exchange must clear, and the clearance itself is characterized in PCR-005, PCR-007 and PCR-008.

Two things this step does not do should be equally clear. The step does not control host cell protein or residual DNA to the drug substance criteria, and makes no viral clearance claim of any kind. The consolidated control strategy, including the cumulative viral clearance and the cross step credit for the shared attributes, is in PCMR-001.

11 Discussion

After both designed experiments, the step is understood well enough to define a design space and to justify the ranges Stage 2 will run. A screening design ranked the 5 factors and a response surface design modelled the region defined by the 4 that screening retained: culture pH, culture temperature, culture duration and dissolved CO₂ (pCO₂). The mechanistic expectations carried into the study were confirmed for every response except acidic variants, and the effects that matter most are the ones prior knowledge predicted would matter.

Confidence in the scale down model is the load bearing assumption of everything above. Reproducibility of the model at the centre point runs from 4.1 % to 9.9 % relative standard deviation, most of which is analytical, and the qualification under SOP-1001 matched the model to at-scale data on the attributes this study measures. However, a scale down model reproduces the response of a culture to the controlled parameters, not the variability a commercial vessel adds. Design spaces are not confirmed at scale at the edges of the ranges, and this one is no exception.

Three limitations qualify the conclusions, and each is managed forward with a named control. The first limitation is the omission of osmolality from the response surface design, where screening resolved a temperature by osmolality interaction on galactosylation of 3.7 percentage points ($p = 0.002$) and this report claims nothing predictive about that term. Osmolality is managed by recipe control to 375 to 425 mOsm and by the WC-CPP classification, and a move of the medium or the feed strategy would require the interaction to be studied properly.

The second limitation is the strength of the lack of fit evidence: all 5 models pass the test, and the test rests on 3 degrees of freedom of pure error. Therefore a passing test bounds the evidence for the model form without establishing it. For this reason the weakest model of the five is acidic variants ($p = 0.15$), which also carries the lowest predicted R^2 at 0.60. The very low criticality of that attribute and the wide acceptance band are what make that acceptable, and the model is used only to bound a range and not to predict a batch.

The third limitation is the univariate treatment of dissolved oxygen, initial viable cell conc., basal medium concentration and nutrient feed-1 volume, since interactions between those four parameters and the five multivariate factors were not studied. The classifications of those parameters in §9 rest on univariate evidence and on the platform record, and a proposed change to two of them together falls outside this report.

The deviations recorded during execution are summarized below (§13). Both were impact assessed before the affected data were used: the dissolved carbon dioxide probe offset of 6.5 mmHg is smaller than the reproducibility of every response it could

affect, and the nutrient feed shortfall of 4.2 % left the delivered volume inside the normal operating range. Neither altered a fitted effect, a classification or a range.

No parameter at this step required the most stringent control class. The five quality linked parameters are all well controlled critical process parameters, because the control system holds each of them inside a range this study proved acceptable and an excursion would be detected in process. The parameter closest to needing more is culture pH, since it moves every response including the one with the tightest capability, and the co-varied proven acceptable range for culture pH in Table 7.1 falls short of the characterization range for every one of them.

12 Conclusions

The production bioreactor is characterized and the operating region is defined, on a response surface model over culture pH, culture temperature, culture duration and dissolved CO₂ (pCO₂) that describes the 5 attributes the step forms at an R² of 91.1% or better and with no significant lack of fit against the centre point replicates.

That operating region is the multivariate space in those four parameters over which every governed attribute stays within the in-process criterion, and it covers 57 % of the characterized grid. The normal operating ranges in Table 4.1 lie inside the region apart from the single corner reported in §7. Proven acceptable ranges are reported per attribute and per parameter in Table 7.1, under both the at set-point and the co-varied analysis.

The attributes the step governs all meet the drug substance criteria at commercial scale, on the estimate in Table 8.1. The tightest capability of the 7 attributes is the one for high mannose, at a Cpk of 1.94 and a margin of 2.99 percentage points to the binding limit. Of the 9 parameters, 5 are well controlled critical process parameters, 3 are key process parameters and the remaining one is a general process parameter, and the step has no critical process parameter.

The 2 recorded deviations were investigated, retained and impact assessed, and neither changed a classification or a range. The capability and the region reported here are estimates from a qualified scale down model, and confirmation at commercial scale is a Stage 2 activity. This report rolls up into PCMR-001.

13 Deviations from the plan

During execution, 2 deviations were recorded against PCP-003. Each was investigated, root caused and impact assessed before the affected data were used, both were retained, and Table 13.1 gives the register.

Table 13.1: Deviations recorded during execution of PCP-003.

Devia- tion	Summary	Detected during	Disposi- tion
DEV-003-01	Dissolved-CO2 probe drift during screening run 4	in-process monitoring vs offline reference	retained
DEV-003-02	Nutrient feed-1 under-delivery on a centre-point batch	post-execution mass reconciliation	retained

Both entries are minor and both were detected by a check that the plan required, one in process and one after execution, and neither invalidated a design or triggered re-execution.

13.1 DEV-003-01 — dissolved carbon dioxide probe drift

The dissolved carbon dioxide probe on the scale down bioreactor read 6.5 mmHg away from the offline reference during screening run 4. The probe drift was detected by the routine comparison of in-process monitoring against the offline reference, and the affected run had already completed when the comparison was made.

The investigation of that drift attributed the offset to probe drift between scheduled calibrations on equipment EQ-BRX-205, which was within its calibration interval at the time of execution. The magnitude of the offset corresponds to 0.11 coded units of the dissolved carbon dioxide factor, or 11 % of the width of the normal operating range, against a characterization range of 40 to 160 mmHg. The actual dissolved carbon dioxide level in the affected run stayed well inside the characterized range, and the run remained a valid setting of the design.

The impact on the fitted effects is bounded by that magnitude. Applied to the largest dissolved carbon dioxide effect in the study, on acidic variants, a misassignment of 0.11 coded units corresponds to a response shift of 0.44 percentage points. The centre point standard deviation of the same response on the screening design is 1.09 percentage points. The offset is smaller than the reproducibility of the measurement it would have to move, and smaller still relative to the glycan responses. The run

was retained, and the probe was recalibrated before the response surface design was executed.

13.2 DEV-003-02 — nutrient feed-1 under-delivery

Nutrient feed-1 was under-delivered by 4.2 % of the scheduled volume on one centre point batch of the screening design, from lot LOT-FED-3120. Post-execution mass reconciliation of that batch found the shortfall, and investigation attributed it to a feed pump calibration offset.

On that batch, the delivered feed volume was 11.5 % of working volume against a set-point of 12 % and a normal operating range of 10 to 14 %. That feed volume kept the batch inside the normal operating range for the parameter throughout, and nutrient feed-1 volume is classified KPP on evidence that it governs titre rather than product quality (§9).

The affected batch is one of the 3 screening centre points. The values of that batch therefore enter the pure error term and not an effect estimate. For the affected batch, the glycan responses were verified on 3 measurements under AMV-3010, whose intermediate precision of 2.5 % gives that verification set a 95 % confidence half width of 6.21 % relative, which puts a feed shortfall of 4.2 % below the level that comparison can resolve. The batch was retained as a centre point, the pump was recalibrated, and the feed delivery check was added to the in-process record for the remaining runs.

Neither deviation changed a parameter classification, a fitted effect or a boundary of the operating region. Both are closed.

14 Appendix A — Screening design matrix and responses

The coded screening matrix and the measured responses are given in Table 14.1. Coded levels are minus one at the low edge of the characterization range, zero at the set-point and plus one at the high edge, with the letters as defined in Table 4.2.

Table 14.1: Screening design matrix (coded) and measured responses.

Run	Type	A	B	C	D	E	Afuco- sylation	Galacto- sylation	High man- nose	Acidic variants	Aggre- gates (HMW)
1	fac- to- rial	-1	-1	-1	-1	1	5.18	27.9	6.61	26.4	1.3
2	fac- to- rial	-1	-1	-1	1	-1	8.93	36.1	6.44	26.5	0.783
3	fac- to- rial	-1	-1	1	-1	-1	11.8	34.6	5.63	22.3	0.785
4	fac- to- rial	-1	-1	1	1	1	4.83	14.7	5.46	17.9	1.48
5	fac- to- rial	-1	1	-1	-1	-1	9.72	41.4	4.61	27	0.859
6	fac- to- rial	-1	1	-1	1	1	4.68	27.3	4.68	27.6	1.26
7	fac- to- rial	-1	1	1	-1	1	6.22	14.5	4.65	23.1	1.48
8	fac- to- rial	-1	1	1	1	-1	9.83	31.2	4.45	19.3	0.948
9	fac- to- rial	1	-1	-1	-1	-1	9.61	28	7.77	24.1	1.51

Run	Type	A	B	C	D	E	Afucosylation	Galactosylation	High mannose	Acidic variants	Aggregates (HMW)
10	factorial	1	-1	-1	1	1	7.51	17	7.89	20.4	2.26
11	factorial	1	-1	1	-1	1	7.13	19.5	7.34	12.6	2.53
12	factorial	1	-1	1	1	-1	5.29	17.6	7.01	11.3	1.61
13	factorial	1	1	-1	-1	1	6.74	28.3	6.43	29.6	2.18
14	factorial	1	1	-1	1	-1	5.4	26.7	5.94	24.4	1.62
15	factorial	1	1	1	-1	-1	3.82	25.2	6.1	18.4	1.8
16	factorial	1	1	1	1	1	3.08	29.1	6.15	15.8	2.05
17	center	0	0	0	0	0	7.8	26.4	5.88	21.9	1.64
18	center	0	0	0	0	0	7.51	27.4	5.84	20	1.63
19	center	0	0	0	0	0	8.48	27	7.11	21.8	1.5

15 Appendix B — Response-surface design matrix and responses

The coded response-surface matrix and the measured responses are given in Table 15.1. The design is face centred, so the axial runs sit at plus or minus one on one factor with the others at zero.

Table 15.1: Response-surface design matrix (coded) and measured responses.

Run	Type	A	B	E	C	Afucosylation	Galactosylation	High man-nose	Acidic variants	Aggregates (HMW)
1	factorial	-1	-1	-1	-1	9.43	38.8	5.44	29.8	0.827
2	factorial	-1	-1	-1	1	11.8	29	5.55	17.6	0.791
3	factorial	-1	-1	1	-1	4.5	25.9	5.65	25.1	1.33
4	factorial	-1	-1	1	1	5.38	16.6	5.38	20	1.37
5	factorial	-1	1	-1	-1	10.4	40.5	4.23	25.6	0.803
6	factorial	-1	1	-1	1	10.1	30.7	4.17	19.8	0.707
7	factorial	-1	1	1	-1	5.12	27.1	4.63	30.3	1.24
8	factorial	-1	1	1	1	6.13	17.4	4.64	21.8	1.41
9	factorial	1	-1	-1	-1	8.92	22	6.79	22.9	1.5
10	factorial	1	-1	-1	1	5.89	18.7	8.81	13.6	1.53
11	factorial	1	-1	1	-1	8.05	18.7	6.11	19.3	1.99
12	factorial	1	-1	1	1	5.17	18.7	7.34	12.3	2.15
13	factorial	1	1	-1	-1	5.41	28.7	6.97	28.5	1.45
14	factorial	1	1	-1	1	2.84	29.1	7.36	18.5	1.7

Run	Type	A	B	E	C	Afucosy- lation	Galacto- sylation	High man- nose	Acidic variants	Aggre- gates (HMW)
15	facto- rial	1	1	1	-1	6.2	25.4	5.95	26.5	2.08
16	facto- rial	1	1	1	1	4.19	27.2	6.8	17.2	2.01
17	axial	-1	0	0	0	7.82	26.1	4.59	24.2	1.15
18	axial	1	0	0	0	6.07	23.7	7.72	19.3	1.82
19	axial	0	-1	0	0	8.77	24.3	6.75	19.9	1.51
20	axial	0	1	0	0	7.55	27.9	5.51	26.8	1.32
21	axial	0	0	-1	0	10.3	30.8	6.11	20.8	1.3
22	axial	0	0	1	0	6.89	22.5	5.54	22.4	2.06
23	axial	0	0	0	-1	8.48	27.1	6.39	25.7	1.5
24	axial	0	0	0	1	8.39	23.5	5.94	19.2	1.45
25	cen- ter	0	0	0	0	7.88	23	5.75	20.5	1.52
26	cen- ter	0	0	0	0	7.95	27.3	6.32	22	1.6
27	cen- ter	0	0	0	0	7.35	25	5.6	22.4	1.55
28	cen- ter	0	0	0	0	7.27	28.2	4.96	21.1	1.37

16 Appendix C — Full effect and coefficient tables

Full screening effect tables for all responses follow, in the order the responses are measured. Effects are twice the coded coefficient.

Table 16.1: Screening effects — Afucosylation.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
AE	2.46	1.23	0.266	4.62	0.0191	*
E	-2.37	-1.19	0.266	-4.46	0.0209	*
AC	-1.76	-0.88	0.266	-3.31	0.0455	*
A	-1.57	-0.785	0.266	-2.95	0.06	
B	-1.34	-0.671	0.266	-2.53	0.0858	
D	-1.33	-0.664	0.266	-2.5	0.0878	
AB	-1.28	-0.64	0.266	-2.41	0.0952	
C	-0.73	-0.365	0.266	-1.37	0.264	
BD	0.452	0.226	0.266	0.85	0.458	
BE	0.362	0.181	0.266	0.681	0.545	
AD	-0.175	-0.0876	0.266	-0.329	0.764	
BC	-0.171	-0.0856	0.266	-0.322	0.769	
CD	-0.147	-0.0736	0.266	-0.277	0.8	
DE	0.0358	0.0179	0.266	0.0674	0.951	
CE	0.0118	0.00591	0.266	0.0222	0.984	

Table 16.2: Screening effects — Galactosylation.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
E	-7.82	-3.91	0.193	-20.2	0.000264	***
AE	6.9	3.45	0.193	17.9	0.000382	***
C	-5.79	-2.9	0.193	-15	0.000644	***
A	-4.56	-2.28	0.193	-11.8	0.00131	**
BD	3.7	1.85	0.193	9.57	0.00242	**
AC	3.6	1.8	0.193	9.32	0.00261	**
B	3.54	1.77	0.193	9.17	0.00274	**
AB	3.28	1.64	0.193	8.49	0.00343	**
D	-2.45	-1.23	0.193	-6.35	0.0079	**
CD	2.18	1.09	0.193	5.64	0.011	*
DE	1.93	0.963	0.193	4.98	0.0155	*

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
BE	1.49	0.747	0.193	3.87	0.0306	*
AD	-0.195	-0.0974	0.193	-0.504	0.649	
BC	-0.131	-0.0657	0.193	-0.34	0.756	
CE	0.119	0.0596	0.193	0.309	0.778	

Table 16.3: Screening effects — High mannose.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	1.51	0.756	0.155	4.89	0.0164	*
B	-1.39	-0.696	0.155	-4.5	0.0205	*
C	-0.446	-0.223	0.155	-1.44	0.245	
BC	0.369	0.185	0.155	1.19	0.318	
E	0.158	0.0791	0.155	0.511	0.645	
D	-0.139	-0.0693	0.155	-0.448	0.685	
AC	0.0897	0.0449	0.155	0.29	0.791	
AE	0.0894	0.0447	0.155	0.289	0.792	
DE	-0.0718	-0.0359	0.155	-0.232	0.831	
CE	-0.0552	-0.0276	0.155	-0.179	0.87	
BE	0.046	0.023	0.155	0.149	0.891	
AB	0.0432	0.0216	0.155	0.14	0.898	
AD	-0.0239	-0.012	0.155	-0.0774	0.943	
CD	-0.0232	-0.0116	0.155	-0.075	0.945	
BD	-0.000576	-0.000288	0.155	-0.00186	0.999	

Table 16.4: Screening effects — Acidic variants.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
C	-8.15	-4.08	0.239	-17	0.000441	***
A	-4.17	-2.08	0.239	-8.7	0.00319	**
B	2.97	1.48	0.239	6.19	0.00848	**
D	-2.55	-1.28	0.239	-5.34	0.0129	*
AB	1.98	0.992	0.239	4.15	0.0255	*
AC	-1.93	-0.964	0.239	-4.03	0.0275	*
BE	1.73	0.865	0.239	3.61	0.0364	*
AD	-0.669	-0.335	0.239	-1.4	0.256	
CE	-0.494	-0.247	0.239	-1.03	0.378	
CD	-0.492	-0.246	0.239	-1.03	0.379	
BD	-0.215	-0.108	0.239	-0.449	0.684	
BC	0.151	0.0757	0.239	0.316	0.773	
DE	0.0324	0.0162	0.239	0.0676	0.95	
E	0.0235	0.0117	0.239	0.049	0.964	
AE	0.0226	0.0113	0.239	0.0472	0.965	

Table 16.5: Screening effects — Aggregates (HMW).

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	0.834	0.417	0.0207	20.1	0.000268	***
E	0.578	0.289	0.0207	13.9	0.0008	***
BE	-0.141	-0.0705	0.0207	-3.4	0.0424	*
C	0.116	0.058	0.0207	2.8	0.0678	
CD	-0.0742	-0.0371	0.0207	-1.79	0.171	
AD	-0.0666	-0.0333	0.0207	-1.61	0.206	
AB	-0.0593	-0.0297	0.0207	-1.43	0.248	
DE	-0.0554	-0.0277	0.0207	-1.34	0.274	
D	-0.0548	-0.0274	0.0207	-1.32	0.278	
BD	-0.0547	-0.0274	0.0207	-1.32	0.278	
AE	0.0416	0.0208	0.0207	1	0.39	
BC	-0.0235	-0.0118	0.0207	-0.567	0.61	
CE	0.0222	0.0111	0.0207	0.537	0.629	
B	-0.00671	-0.00336	0.0207	-0.162	0.882	
AC	-0.00663	-0.00332	0.0207	-0.16	0.883	

Full response-surface coefficient tables for the responses not shown in §5.3 follow.

Table 16.6: Response-surface coefficients — Afucosylation.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	8.04	0.202	39.8	5.74e-15	***
A	-0.997	0.138	-7.24	6.52e-06	***
B	-0.551	0.138	-4	0.00151	**
E	-1.3	0.138	-9.45	3.45e-07	***
C	-0.367	0.138	-2.67	0.0194	*
AB	-0.63	0.146	-4.32	0.000837	***
AE	1.32	0.146	9.06	5.6e-07	***
AC	-0.907	0.146	-6.21	3.16e-05	***
BE	0.362	0.146	2.48	0.0277	*
BC	-0.0703	0.146	-0.481	0.638	
EC	0.0322	0.146	0.221	0.829	
A ²	-1.37	0.364	-3.77	0.00233	**
B ²	-0.157	0.364	-0.431	0.673	
E ²	0.257	0.364	0.705	0.493	
C ²	0.117	0.364	0.323	0.752	

Table 16.7: Response-surface coefficients — Acidic variants.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	22	0.542	40.6	4.43e-15	***

Term	Coef.	Std. err.	t	p-value	Sig.
A	-2.01	0.369	-5.44	0.000113	***
B	1.9	0.369	5.14	0.000191	***
E	-0.124	0.369	-0.335	0.743	
C	-4.09	0.369	-11.1	5.4e-08	***
AB	1.11	0.392	2.84	0.0139	*
AE	-0.793	0.392	-2.03	0.0638	
AC	-0.248	0.392	-0.633	0.538	
BE	0.655	0.392	1.67	0.118	
BC	-0.0111	0.392	-0.0283	0.978	
EC	0.465	0.392	1.19	0.256	
A ²	-0.586	0.975	-0.601	0.558	
B ²	1.03	0.975	1.06	0.311	
E ²	-0.718	0.975	-0.736	0.475	
C ²	0.13	0.975	0.133	0.896	

Table 16.8: Response-surface coefficients — Aggregates (HMW).

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	1.52	0.0341	44.8	1.25e-15	***
A	0.367	0.0232	15.8	7.26e-10	***
B	-0.0157	0.0232	-0.675	0.512	
E	0.281	0.0232	12.1	1.91e-08	***
C	0.0225	0.0232	0.971	0.349	
AB	0.0135	0.0246	0.547	0.593	
AE	-0.0106	0.0246	-0.43	0.674	
AC	0.0179	0.0246	0.727	0.48	
BE	-0.00713	0.0246	-0.29	0.777	
BC	0.00414	0.0246	0.168	0.869	
EC	0.0105	0.0246	0.426	0.677	
A ²	-0.0533	0.0613	-0.87	0.4	
B ²	-0.119	0.0613	-1.93	0.0752	
E ²	0.143	0.0613	2.33	0.0366	*
C ²	-0.0626	0.0613	-1.02	0.326	

17 Appendix D — Analytical methods summary

The validated methods used for the designed responses and for the impurity attributes are summarized in Table 17.1. Method performance is reported in full in the validation reports named in the table.

Table 17.1: Validated analytical methods and the attributes they report.

Method validation	Title	Attribute reported	Reportable unit
AMV-3010	N-Glycan Map (2-AB HILIC-UPLC)	Afucosylation; Galactosylation (total %Gal); High Mannose	% of glycans; % (G1+G2 equiv.)
AMV-3011	Size-Variants (SEC-HPLC)	Aggregates (HMW)	% HMW (SEC)
AMV-3012	Host-Cell Protein ELISA	Host Cell Protein (HCP)	ng/mg
AMV-3013	Charge Variants (icIEF)	Acidic charge variants (deamidation)	% (CEX/icIEF)
AMV-3014	Residual DNA (qPCR)	Residual DNA	ng/dose

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